

小小爆米花厂

1.0.0 程序鉴别材料

内部草稿 / 待权利人确认

本封面为内部审阅说明，不计入后续程序页码。

本次按真实源文件顺序汇集9个文件，全量排版34页。

程序正文交存方式草案：不足60页，全部列出。

正文每页50个排版行，末页按实际剩余内容，不增加空代码凑页。

长源行按版宽折行，左侧为文件编号:原始行号，+表示该行续排。

页眉标明软件名称、版本和程序页码；原始源文件及SHA-256另附。

空行来自源文件，未另增凑页；折行不是新增逻辑代码。

本材料不含本地配置、真实广告ID、生成bundle或第三方库。

权利人、开发日期、开发方式及申请表事实待本人确认；未填写登记号。

提交前须核对受理端对行数、折行、封面及文件格式的具体要求。

内部草稿应核对后再用于正式登记，不能视为已登记证明。

文件编号与正文来源

01	src/main.js	196	原始行
02	src/core.js	305	原始行
03	src/platform.js	430	原始行
04	src/renderer.js	145	原始行
05	src/audio.js	112	原始行
06	tools/build.mjs	37	原始行
07	tools/audio.mjs	19	原始行
08	tools/serve.mjs	16	原始行
09	web/index.html	24	原始行

src/main.js:1 - src/main.js:39

```
01:0001 'use strict';
01:0002 const { Game, CONFIG, formatNumber } = require('./core');
01:0003 const { createPlatform } = require('./platform');
01:0004 const { AudioEngine } = require('./audio');
01:0005 const { Renderer } = require('./renderer');
01:0006
01:0007 const platform = createPlatform();
01:0008 const canvas = platform.canvas;
01:0009 const ctx = canvas.getContext('2d');
01:0010 const sound = new AudioEngine();
01:0011 const initialSave=platform.load();
01:0012 let initialLoadError=platform.lastStorageError;
01:0013 let game = new Game({ save: initialSave });
01:0014 const renderer = new Renderer(ctx);
01:0015 const ui = { tab:'upgrades', modal:null, toast:'', toastSeconds:0, adBusy:false, saved:false,
01:0015+ isDouyin:platform.isDouyin,
01:0016   startup:true, sidebarBusy:false, sidebar:{supported:false,checking:false,fromSidebar:false} };
01:0017 // Only display real identifiers explicitly supplied in the local release configuration.
01:0018 const publication=platform.config&&platform.config.publication||{};
01:0019 ui.publication=[[ '批准文号',publication.approvalNumber],[ '网络游戏出版号',
01:0019+ publication.publicationNumber],[ '著作权登记号',publication.copyrightNumber]]
01:0020   .filter(([ ,value])=>typeof value==='string'&&value.trim()).map(([label,value])=>`${label}: ${
01:0020+ value.trim().slice(0,100)}`);
01:0021 let scale=1, ox=0, oy=0, ratio=1, width=480, height=920;
01:0022 let hidden=false, hideForAd=false, lastTime=0, saveTimer=0, autoParticleTimer=0,
01:0022+ toastSaveFailed=false, pointer=null;
01:0023 const reasons={
01:0024   'not-enough-coins':'金币还差一点, 继续生产吧', 'orders-required':'先完成所需主线订单',
01:0024+ 'max-level':'这项升级已经满级',
01:0025   'max-machine':'全部设备已经落成', 'order-not-ready':'再生产一些, 就能装车了',
01:0025+ 'intro-first':'先熟悉工厂, 营业 90 秒后开放激励',
01:0026   'reward-cooldown':'激励正在准备, 请稍后再来', 'no-offline-reward':'离线收益已经领取',
01:0026+ 'already-affordable':'金币足够, 可以直接换代',
01:0027   'production-required':'先提升自动生产能力', 'already-claimed':'这份奖励已经到账',
01:0027+ 'stale-order':'订单已结算, 奖励不会重复领取',
01:0028   'stale-offline':'离线收益已变化, 请重新查看', 'unknown-reward':'这次奖励已失效, 请重新选择',
01:0028+ 'busy':'广告正在处理中',
01:0029   'unavailable':'广告暂不可用, 工厂仍可正常生产', 'cancelled':'未完整观看, 未发放广告奖励',
01:0029+ 'failed':'广告加载失败, 请稍后再试',
01:0030   'timeout':'广告响应超时, 工厂可继续营业'
01:0031 };
01:0032 function toast(message) {ui.toast=message;ui.toastSeconds=3.5;}
01:0033 function save() {
01:0034   const ok=platform.save(game.exportSave());ui.saved=ok;
01:0035   if(!ok&&!toastSaveFailed){toast('当前进度暂未保存, 请检查设备存储空间');toastSaveFailed=true;}
01:0036   if(ok)toastSaveFailed=false;
01:0037   return ok;
01:0038 }
01:0039 function closeModal() {if(ui.adBusy)return;if(ui.modal&&ui.modal.quote)game.cancelReward(ui.modal.quote.id);ui.modal=null;save();}
```

src/main.js:40 - src/main.js:86

```

01:0040 function openModal(type) {if(ui.adBusy)return;closeModal();ui.modal={type};sound.play('click');}
01:0041 function configure() {sound.setEnabled(game.state.settings.sound);}
01:0042 function restore() {
01:0043     configure();
01:0044     // Pending quotes survive snapshots only to guarantee unique identifiers. A restarted app
01:0045     // has no verified SDK completion callback, so it must discard unfulfilled requests.
01:0046     for(const id of Object.keys(game.state.pendingRewards))game.cancelReward(id);
01:0047     ui.modal=game.state.offline&&game.state.offline.coins>0?{type:'offline':null;
01:0048     if(game.loadWarning)toast(game.loadWarning);
01:0049     else if(initialLoadError){toast('存档读取失败, 已安全开始新工厂');initialLoadError='';}
01:0050     save();
01:0051 }
01:0052 function resize(info) {
01:0053     width=info.width;height=info.height;ratio=Math.min(2,info.pixelRatio||1);
01:0054     canvas.width=Math.round(width*ratio);canvas.height=Math.round(height*ratio);
01:0055     const safe=info.safeArea||{};
01:0056     const top=platform.isDouyin?Math.max(0,safe.top||0)+18:0;
01:0057     const bottom=platform.isDouyin?Math.max(0,height-(safe.bottom||height))+6:0;
01:0058     scale=Math.min(width/480,(height-top-bottom)/920);ox=(width-480*scale)/2;
01:0058+ oy=top+(height-top-bottom-920*scale)/2;
01:0059 }
01:0060 function processEvents() {
01:0061     for(const e of game.drainEvents()){
01:0062         if(e.type==='produce'){
01:0063             if(e.source==='tap'){renderer.emit(e);sound.play('pop');}
01:0064             else if(e.source==='auto'&&autoParticleTimer>=.12){renderer.emit(e);autoParticleTimer=0;}
01:0065         }else{
01:0066             renderer.emit(e);sound.play(e.type==='evolve'? 'machine':e.type);
01:0067             platform.track(e.type,e);
01:0068             if(e.type==='burst'&&game.state.settings.haptics)platform.vibrate();
01:0069             if(e.type==='evolve')toast(`${game.getView().machine.name}开动了!`);
01:0070             if(e.type==='complete')ui.modal={type:'completion'};
01:0071         }
01:0072     }
01:0073 }
01:0074 function result(actionResult) {
01:0075     if(!actionResult||!actionResult.ok){if(actionResult)toast(reasons[actionResult.reason]||'暂时无法操作
01:0075+ , 请稍后再试');return false;}
01:0076     processEvents();save();return true;
01:0077 }
01:0078 function offerReward(kind) {
01:0079     const view=game.getView(),offer=view.rewards[kind];
01:0080     if(!offer||!offer.available){toast(offer&&offer.reason==='reward-cooldown'?`下次激励还需 ${
01:0080+ Math.ceil(offer.cooldown)} 秒`:reasons[offer&&offer.reason]||'本次激励暂不可用');return;}
01:0081     closeModal();const quote=game.quoteReward(kind);
01:0082     if(!quote){toast('奖励正在准备, 请稍后再试');return;}
01:0083     ui.modal={type:'reward',quote};platform.track('reward_offer',{kind,amount:quote.amount});save();
01:0084 }
01:0085 async function watch(simulate) {
01:0086     if(ui.adBusy||!ui.modal||!ui.modal.quote)return;

```

src/main.js:87 - src/main.js:128

```

01:0087   const quote=ui.modal.quote;ui.adBusy=true;save();
01:0088   let response;
01:0089   try{response=await platform.reward(quote.kind,simulate?{simulate}:undefined);}catch(_){response={
01:0089+ completed:false,reason:'failed'}};
01:0090   ui.adBusy=false;
01:0091   if(response.completed){
01:0092       ui.modal=null;const applied=game.applyReward(quote.id);
01:0093       if(result(applied)){toast(quote.kind==='turbo'?`涡轮启动! 自动生产 3 倍, 持续 90 秒:`奖励到账 +${
01:0093+ formatNumber(applied.coins)} 金币`);platform.track('reward_granted',{kind:quote.kind,
01:0093+ coins:applied.coins});}
01:0094   }else{
01:0095       game.cancelReward(quote.id);
01:0096       ui.modal=quote.kind==='order'?{type:'order':quote.kind==='offline'&&game.state.offline?{
01:0096+ type:'offline':quote.kind==='sponsor'?{type:'machine':null;
01:0097       toast(reasons[response.reason]||`广告未完成, 奖励未领取`);save();
01:0098   }
01:0099   lastTime=0;
01:0100 }
01:0101 async function maybeInterstitial() {
01:0102     if(ui.adBusy||hidden||ui.modal)return;
01:0103     ui.adBusy=true;
01:0104     try{await platform.interstitial();}finally{ui.adBusy=false;lastTime=0;save();}
01:0105 }
01:0106 async function visitSidebar() {
01:0107     if(ui.sidebarBusy||typeof platform.navigateSidebar!=='function')return;
01:0108     ui.sidebarBusy=true;save();
01:0109     try {
01:0110         // Invoke the native API within this click stack, before the first await.
01:0111         const response=await platform.navigateSidebar();
01:0112         if(!response.ok)toast(response.reason==='checking'?`入口正在准备, 请稍后再试`:`暂时无法打开侧边栏, 请
01:0112+ 稍后再试`);
01:0113     }catch(_){toast(`暂时无法打开侧边栏, 请稍后再试`);}
01:0114     finally{ui.sidebarBusy=false;}
01:0115 }
01:0116 function act(action) {
01:0117     if(!action||ui.adBusy)return;
01:0118     if(ui.startup){
01:0119         if(action==='start'){ui.startup=false;pointer=null;lastTime=0;sound.unlock();save();}
01:0120         return;
01:0121     }
01:0122     sound.unlock();
01:0123     if(action==='tap'){if(!ui.modal){game.tap();processEvents();}return;}
01:0124     if(action.startsWith('tab:')){ui.tab=action.slice(4);sound.play('click');return;}
01:0125     if(action.startsWith('upgrade:')){const key=action.slice(8);if(result(game.buyUpgrade(key)))platform
01:0125+ .track('upgrade_purchase',{kind:key,afterReward:game.state.playedSeconds-game.state.lastRewardAt<60});
01:0125+ return;}
01:0126     if(action.startsWith('ad:')){offerReward(action.slice(3));return;}
01:0127     if(action.startsWith('simulate:')){watch(action.slice(9));return;}
01:0128     if(action.startsWith('setting:')){const key=action.slice(8);game.setSetting(key,
01:0128+ !game.state.settings[key]);configure();if(game.state.settings.sound)sound.play('click');save();return;}

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src/main.js:128 - src/main.js:169

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01:0128+ }
01:0129   if(action==='watch'){watch();return;}
01:0130   if(action==='visitSidebar'){visitSidebar();return;}
01:0131   if(action==='close'){closeModal();return;}
01:0132   if(action==='claimOrder'){ui.modal=null;if(result(game.claimOrder())){toast('订单已发车, 金币到账!');
01:0132+ if(game.state.orderIndex%4===0)maybeInterstitial();}return;}
01:0133   if(action==='claimOffline'){ui.modal=null;if(result(game.claimOffline()))toast('离线收益已领取, 欢迎回
01:0133+ 来!');return;}
01:0134   if(action==='evolve'){ui.modal=null;if(result(game.evolve())){ui.tab='machines';maybeInterstitial();
01:0134+ }return;}
01:0135   if(action==='completion'&&!game.getView().completed){toast('完成 20 个主线订单后, 可获得竣工纪念');
01:0135+ return;}
01:0136   openModal(action);
01:0137 }
01:0138
01:0139 platform.onPointer(e=>{
01:0140   const x=(e.x-ox)/scale,y=(e.y-oy)/scale;
01:0141   if(e.type==='down'){
01:0142     if(pointer)return;
01:0143     const action=renderer.actionAt(x,y);pointer={id:e.id,action,x,y};
01:0144     // Production responds immediately. Reward/navigation buttons activate on release,
01:0145     // so repeated production taps cannot drift into an ad request.
01:0146     if(action==='tap')act(action);
01:0147   }else if((e.type==='up'||e.type==='cancel')&&pointer&&pointer.id===e.id){
01:0148     const p=pointer;pointer=null;
01:0149     if(e.type==='up'&&p.action!=='tap'&&renderer.actionAt(x,y)===p.action&&Math.hypot(x-p.x,
01:0149+ y-p.y)<20)act(p.action);
01:0150   }
01:0151 });
01:0152 platform.onResize(resize);
01:0153 platform.onHide(=>{hideForAd=ui.adBusy;hidden=true;pointer=null;sound.setEnabled(false);save();});
01:0154 platform.onShow(=>{
01:0155   hidden=false;lastTime=0;
01:0156   const settledAdAbsence=hideForAd&&!ui.adBusy&&(Date.now()-game.state.savedAt)>=CONFIG.offlineMinSeco
01:0156+ nds*1000;
01:0157   if(!hideForAd||settledAdAbsence){const snapshot=game.exportSave(game.state.savedAt);game=new Game({
01:0157+ save:snapshot});restore();platform.track('session_resume',{});
01:0158   hideForAd=false;
01:0159   configure();
01:0160 });
01:0161 // The Douyin IDE also exposes browser globals, but tt canvases have no DOM API.
01:0162 if(!platform.isDouyin&&typeof document!=='undefined'){
01:0163   canvas.id='game';canvas.setAttribute('aria-label','小小爆米花厂游戏画面');canvas.setAttribute('role',
01:0163+ 'application');canvas.tabIndex=0;
01:0164   const loading=document.getElementById('loading');if(loading)loading.remove();
01:0165   window.addEventListener('keydown',e=>{
01:0166     if(ui.startup){if(e.code==='Enter'){e.preventDefault();act('start');}return;}
01:0167     if(e.repeat&&e.code!=='Space')return;
01:0168     if(e.code==='Escape'){act('close');return;}
01:0169     if(ui.modal)return;

```

src/main.js:170 - src/core.js:13

```

01:0170     const keys={Space:'tap',Digit1:'upgrade:tap',Digit2:'upgrade:auto',Digit3:'upgrade:value',
01:0170+ Key0:'order',KeyM:'machine'};
01:0171     if(keys[e.code]){e.preventDefault();act(keys[e.code]);}
01:0172   });
01:0173   // Read-only diagnostics for QA; no cheats or developer controls in the player flow.
01:0174   globalThis.__POPCORN__={snapshot:()=>JSON.parse(JSON.stringify(game.getView()))},
01:0174+ analytics:()=>platform.getAnalytics(),version:'1.0.0'};
01:0175 }
01:0176 restore();resize(platform.getSystemInfo());platform.track('session_start',{isDouyin:platform.isDouyin}
01:0176+ );
01:0177 if(typeof platform.checkSidebar==='function')platform.checkSidebar().catch(()=>{});
01:0178 const runtimeGlobal=typeof globalThis!=='undefined'?globalThis:GameGlobal;
01:0179 const requestFrame=typeof requestAnimationFrame==='function'?requestAnimationFrame.bind(runtimeGlobal)
01:0179+ :cb=>setTimeout(()=>cb(Date.now()),1000/30);
01:0180 function frame(time) {
01:0181   const dt=lastTime?Math.max(0,Math.min(60,(time-lastTime)/1000)):0;lastTime=time;
01:0182   if(!hidden){
01:0183     if(!ui.adBusy){
01:0184       if(!ui.startup){game.tick(dt);autoParticleTimer+=dt;processEvents();}
01:0185       saveTimer+=dt;if(saveTimer>=5){saveTimer=0;save();}
01:0186     }
01:0187     if(!ui.startup&&ui.toastSeconds>0){ui.toastSeconds-=dt;if(ui.toastSeconds<=0)ui.toast='';}
01:0188     if(typeof platform.getSidebarState==='function')ui.sidebar=platform.getSidebarState();
01:0189     ctx.setTransform(ratio,0,0,ratio,0,0);ctx.fillStyle='#e9eee5';ctx.fillRect(0,0,width,height);
01:0190     // Subtle vertical rails make the portrait canvas feel intentional on wide screens.
01:0191     if(width>480*scale+60){ctx.strokeStyle='#dce3d7';ctx.lineWidth=1;for(const x of [ox-22,
01:0191+ ox+480*scale+22]){ctx.beginPath();ctx.moveTo(x,0);ctx.lineTo(x,height);ctx.stroke();}}
01:0192     ctx.setTransform(ratio*scale,0,0,ratio*scale,ratio*ox,ratio*oy);renderer.draw(game.getView(),ui,
01:0192+ Math.min(.05,dt));
01:0193   }
01:0194   requestFrame(frame);
01:0195 }
01:0196 requestFrame(frame);
02:0001 'use strict';
02:0002 // Platform-independent simulation. Tick uses seconds; save timestamps use milliseconds.
02:0003 const CONFIG = {
02:0004   version: 1, title: '小小爆米花厂', energyMax: 100, tapEnergy: 2, passiveEnergy: 1,
02:0005   maxUpgradeLevel: 24, autoLevelGrowth: 1.3, turboMultiplier: 3, turboDuration: 90,
02:0006   rewardUnlockSeconds: 90, rewardCooldownSeconds: 120,
02:0007   offlineMaxSeconds: 28800, offlineEfficiency: 0.5, offlineMinSeconds: 30,
02:0008   machines: [
02:0009     { id: 0, name: '手摇锅', description: '一把玉米, 开启你的工厂梦', color: '#f3b35b', multiplier: 1,
02:0009+ priceMultiplier: 1, cost: 0, requiredOrders: 0 },
02:0010     { id: 1, name: '电热锅', description: '恒温加热, 香气开始飘满街', color: '#ed785d', multiplier: 2.4,
02:0010+ priceMultiplier: 1.12, cost: 30000, requiredOrders: 3 },
02:0011     { id: 2, name: '双缸机', description: '双锅轮转, 让每一份等待更短', color: '#66bfaa', multiplier:
02:0011+ 5.5, priceMultiplier: 1.25, cost: 2000000, requiredOrders: 6 },
02:0012     { id: 3, name: '多头机', description: '六头齐开, 爆米花像瀑布落下', color: '#6baacb', multiplier:
02:0012+ 12, priceMultiplier: 1.4, cost: 160000000, requiredOrders: 10 },
02:0013     { id: 4, name: '自动流水线', description: '连续生产, 整座工厂为你运转', color: '#9c86cf',

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src/core.js:13 - src/core.js:60

```

02:0013+ multiplier: 26, priceMultiplier: 1.6, cost: 10000000000, requiredOrders: 14 },
02:0014   { id: 5, name: '巨型爆米花塔', description: '让金色爆米花, 点亮整座城市', color: '#e5ad4d',
02:0014+ multiplier: 60, priceMultiplier: 1.8, cost: 225000000000, requiredOrders: 18 }
02:0015 ],
02:0016 upgrades: {
02:0017   tap: { name: '爆裂玉米', description: '每次点击产量提升', baseCost: 16, growth: 2.1 },
02:0018   auto: { name: '自动火力', description: '持续自动生产并出售', baseCost: 24, growth: 2.1 },
02:0019   value: { name: '焦糖配方', description: '所有爆米花的售价提升', baseCost: 32, growth: 2.1 }
02:0020 },
02:0021 orders: [
02:0022   { name: '街角第一桶', target: 50, reward: 180 },
02:0023   { name: '放学后的香气', target: 200, reward: 600 },
02:0024   { name: '周末小摊', target: 800, reward: 2600 },
02:0025   { name: '社区电影夜', target: 2500, reward: 8800 },
02:0026   { name: '甜蜜下午茶', target: 7000, reward: 32000 },
02:0027   { name: '校园运动会', target: 20000, reward: 120000 },
02:0028   { name: '双倍快乐', target: 60000, reward: 480000 },
02:0029   { name: '公园嘉年华', target: 150000, reward: 1440000 },
02:0030   { name: '满座电影院', target: 350000, reward: 4400000 },
02:0031   { name: '全城首映礼', target: 800000, reward: 12800000 },
02:0032   { name: '星光音乐节', target: 7200000, reward: 40000000 },
02:0033   { name: '金色美食街', target: 16000000, reward: 112000000 },
02:0034   { name: '跨城快递', target: 32000000, reward: 288000000 },
02:0035   { name: '连锁店开张', target: 64000000, reward: 720000000 },
02:0036   { name: '超级体育场', target: 120000000, reward: 1800000000 },
02:0037   { name: '海滨烟花夜', target: 220000000, reward: 4400000000 },
02:0038   { name: '全国巡游', target: 400000000, reward: 10400000000 },
02:0039   { name: '城市庆典', target: 720000000, reward: 24000000000 },
02:0040   { name: '云端甜品节', target: 1200000000, reward: 52000000000 },
02:0041   { name: '世界爆米花日', target: 2000000000, reward: 100000000000 }
02:0042 ]
02:0043 };
02:0044 const MAX_NUMBER = 1e150;
02:0045 const KEYS = ['tap', 'auto', 'value'];
02:0046 const REWARDS = ['turbo', 'order', 'sponsor', 'offline'];
02:0047 const finite = (n, fallback = 0, max = MAX_NUMBER) => typeof n === 'number' && Number.isFinite(n) ?
02:0047+ Math.max(0, Math.min(max, n)) : fallback;
02:0048 const integer = (n, fallback = 0, max = MAX_NUMBER) => Math.floor(finite(n, fallback, max));
02:0049 const clone = obj => JSON.parse(JSON.stringify(obj));
02:0050 const fail = reason => ({ ok: false, reason });
02:0051 const has = (o, k) => Object.prototype.hasOwnProperty.call(o, k);
02:0052 function formatNumber(value) {
02:0053   const n = finite(value);
02:0054   const trim = x => x.replace(/\.0+$/, '').replace(/\.$/, '');
02:0055   if (n < 10000) return Math.floor(n).toLocaleString('zh-CN');
02:0056   if (n < 100000000) return trim((n / 10000).toFixed(n < 100000 ? 2 : 1)) + '万';
02:0057   if (n < 1e12) return trim((n / 1e8).toFixed(n < 1e9 ? 2 : 1)) + '亿';
02:0058   return trim((n / 1e12).toFixed(1)) + '万亿';
02:0059 }
02:0060 function freshState(now) {

```

src/core.js:61 - src/core.js:101

```

02:0061   return {
02:0062     version: 1, savedAt: now, coins: 0, totalProduced: 0, totalCoins: 0,
02:0063     taps: 0, bursts: 0, machine: 0, upgrades: { tap: 0, auto: 0, value: 0 },
02:0064     orderIndex: 0, loopIndex: 0, energy: 0, boostSeconds: 0, playedSeconds: 0,
02:0065     settings: { sound: true, haptics: true }, offline: null, rewardSerial: 0,
02:0066     rewardedCount: 0, lastRewardAt: -CONFIG.rewardCooldownSeconds,
02:0067     pendingRewards: {}, claimedRewards: [], completedAt: null
02:0068   };
02:0069 }
02:0070 class Game {
02:0071   constructor({ save = null, now = Date.now() } = {}) {
02:0072     this.events = []; this.now = finite(now, Date.now()); this.state = freshState(this.now);
02:0072+ this.loadWarning = null;
02:0073     if (save) this._restore(save);
02:0074   }
02:0075   _restore(input) {
02:0076     let save;
02:0077     try { save = typeof input === 'string' ? JSON.parse(input) : input; }
02:0078     catch (_) { this.loadWarning = '存档读取失败, 已安全开始新工厂'; return; }
02:0079     if (!save || typeof save !== 'object' || Array.isArray(save) || save.version !== CONFIG.version) {
02:0080       this.loadWarning = '存档版本不兼容, 已安全开始新工厂'; return;
02:0081     }
02:0082     const s = this.state;
02:0083     for (const key of ['coins', 'totalProduced', 'totalCoins', 'playedSeconds']) s[key] =
02:0083+ finite(save[key]);
02:0084     for (const key of ['taps', 'bursts', 'loopIndex', 'rewardSerial', 'rewardedCount']) s[key] =
02:0084+ integer(save[key]);
02:0085     s.totalCoins = Math.max(s.totalCoins, s.coins);
02:0086     s.machine = integer(save.machine, 0, 5); s.orderIndex = integer(save.orderIndex, 0, 20);
02:0087     while (s.orderIndex > 0 && CONFIG.orders[s.orderIndex - 1].target > s.totalProduced)
02:0087+ s.orderIndex--;
02:0088     if (s.orderIndex < 20) s.loopIndex = 0;
02:0089     while (s.machine > 0 && CONFIG.machines[s.machine].requiredOrders > s.orderIndex) s.machine--;
02:0090     for (const key of KEYS) s.upgrades[key] = integer(save.upgrades && save.upgrades[key], 0,
02:0090+ CONFIG.maxUpgradeLevel);
02:0091     s.energy = finite(save.energy, 0, CONFIG.energyMax - 1e-10);
02:0092     s.boostSeconds = finite(save.boostSeconds, 0, 86400);
02:0093     s.lastRewardAt = typeof save.lastRewardAt === 'number' && Number.isFinite(save.lastRewardAt)
02:0094     ? Math.min(s.playedSeconds, Math.max(-CONFIG.rewardCooldownSeconds, save.lastRewardAt)) :
02:0094+ -CONFIG.rewardCooldownSeconds;
02:0095     for (const key of ['sound', 'haptics']) if (save.settings && typeof save.settings[key] ===
02:0095+ 'boolean') s.settings[key] = save.settings[key];
02:0096     s.claimedRewards = Array.isArray(save.claimedRewards) ? save.claimedRewards.filter(x => typeof x
02:0096+ === 'string' && x.length < 100).slice(-128) : [];
02:0097     if (s.orderIndex === 20) s.completedAt = finite(save.completedAt, s.playedSeconds);
02:0098     if (save.offline && typeof save.offline.id === 'string') {
02:0099       const o = save.offline;
02:0100       s.offline = { id: o.id.slice(0, 100), seconds: finite(o.seconds, 0, CONFIG.offlineMaxSeconds),
02:0100+ production: finite(o.production), coins: finite(o.coins) };
02:0101   }

```


src/core.js:102 - src/core.js:140

```

02:0102     if (save.pendingRewards && typeof save.pendingRewards === 'object') {
02:0103         for (const [id, q] of Object.entries(save.pendingRewards).slice(-4)) {
02:0104             if (!q || typeof q !== 'object' || id !== q.id || !/^reward:\d+:\d+$/ .test(id) || id.length >
02:0104+ 100 || !REWARDS.includes(q.kind) || s.claimedRewards.includes(id)) continue;
02:0105             s.pendingRewards[id] = {
02:0106                 id, kind: q.kind, title: String(q.title || '').slice(0, 100), description:
02:0106+ String(q.description || '').slice(0, 100), amount: finite(q.amount),
02:0107                 duration: finite(q.duration, 0, CONFIG.turboDuration), orderIndex: integer(q.orderIndex),
02:0107+ loopIndex: integer(q.loopIndex),
02:0108                 offlineId: typeof q.offlineId === 'string' ? q.offlineId.slice(0, 100) : null, machine:
02:0108+ integer(q.machine, 0, 5), createdAt: finite(q.createdAt)
02:0109             };
02:0110         }
02:0111     }
02:0112     const savedAt = finite(save.savedAt, this.now);
02:0113     const elapsed = Math.max(0, (this.now - savedAt) / 1000);
02:0114     s.boostSeconds = Math.max(0, s.boostSeconds - elapsed);
02:0115     s.lastRewardAt = Math.max(-CONFIG.rewardCooldownSeconds, s.lastRewardAt - elapsed);
02:0116     if (elapsed >= CONFIG.offlineMinSeconds) {
02:0117         const seconds = Math.min(elapsed, CONFIG.offlineMaxSeconds - (s.offline ? s.offline.seconds :
02:0117+ 0));
02:0118         if (seconds > 0) {
02:0119             const p = this._production(), prior = s.offline;
02:0120             const production = p.baseAuto * seconds * CONFIG.offlineEfficiency;
02:0121             s.offline = {
02:0122                 id: 'offline:' + savedAt + ':' + Math.floor(this.now), seconds: seconds + (prior ?
02:0122+ prior.seconds : 0),
02:0123                 production: production + (prior ? prior.production : 0), coins: production * p.price +
02:0123+ (prior ? prior.coins : 0)
02:0124             };
02:0125             for (const [id, q] of Object.entries(s.pendingRewards)) if (q.kind === 'offline') delete
02:0125+ s.pendingRewards[id];
02:0126         }
02:0127     }
02:0128     s.savedAt = this.now;
02:0129 }
02:0130 _production() {
02:0131     const s = this.state, m = CONFIG.machines[s.machine];
02:0132     const tap = (1 + 0.8 * s.upgrades.tap) * Math.pow(1.2, s.upgrades.tap) * m.multiplier;
02:0133     const baseAuto = (0.4 + 0.7 * s.upgrades.auto) * Math.pow(CONFIG.autoLevelGrowth,
02:0133+ s.upgrades.auto) * m.multiplier;
02:0134     const price = (1 + 0.2 * s.upgrades.value) * Math.pow(1.15, s.upgrades.value) * m.priceMultiplier;
02:0135     return { tap, auto: baseAuto * (s.boostSeconds > 0 ? 3 : 1), baseAuto, price, baseIncome:
02:0135+ baseAuto * price, turboMultiplier: s.boostSeconds > 0 ? 3 : 1 };
02:0136 }
02:0137 _addCoins(coins) { this.state.coins = finite(this.state.coins + coins); this.state.totalCoins =
02:0137+ finite(this.state.totalCoins + coins); }
02:0138 _produce(amount, source) {
02:0139     const n = finite(amount); if (!n) return;
02:0140     const coins = finite(n * this._production().price);

```

src/core.js:141 - src/core.js:184

```

02:0141     this.state.totalProduced = finite(this.state.totalProduced + n); this._addCoins(coins);
02:0142     this._event('produce', { source, amount: n, coins });
02:0143 }
02:0144 _event(type, extra = {}) {
02:0145     if (this.events.length >= 160) this.events.shift();
02:0146     this.events.push({ type, ...extra });
02:0147 }
02:0148 _addEnergy(amount) {
02:0149     this.state.energy += amount;
02:0150     while (this.state.energy >= CONFIG.energyMax) {
02:0151         this.state.energy -= CONFIG.energyMax;
02:0152         const p = this._production(), production = p.tap * 24 + p.baseAuto * 8;
02:0153         this._produce(production, 'burst'); this.state.bursts++; this._event('burst', { amount:
02:0153+ production });
02:0154     }
02:0155 }
02:0156 tick(dt) {
02:0157     if (typeof dt !== 'number' || !Number.isFinite(dt) || dt <= 0) return fail('invalid-time');
02:0158     dt = Math.min(dt, 60);
02:0159     const p = this._production(), boostedTime = Math.min(dt, this.state.boostSeconds);
02:0160     this._produce(p.baseAuto * (dt + boostedTime * 2), 'auto');
02:0161     this.state.boostSeconds = Math.max(0, this.state.boostSeconds - dt); this.state.playedSeconds +=
02:0161+ dt;
02:0162     this._addEnergy(dt * CONFIG.passiveEnergy); return { ok: true, seconds: dt };
02:0163 }
02:0164 tap() {
02:0165     const amount = this._production().tap;
02:0166     this.state.taps++; this._produce(amount, 'tap'); this._addEnergy(CONFIG.tapEnergy); return { ok:
02:0166+ true, amount };
02:0167 }
02:0168 _upgradeCost(key) {
02:0169     const u = CONFIG.upgrades[key];
02:0170     return Math.ceil(u.baseCost * Math.pow(u.growth, this.state.upgrades[key]) * (1 +
02:0170+ this.state.machine * 0.45));
02:0171 }
02:0172 buyUpgrade(key) {
02:0173     if (!KEYS.includes(key)) return fail('invalid-upgrade');
02:0174     if (this.state.upgrades[key] >= CONFIG.maxUpgradeLevel) return fail('max-level');
02:0175     const cost = this._upgradeCost(key); if (this.state.coins < cost) return fail('not-enough-coins');
02:0176     this.state.coins -= cost; this.state.upgrades[key]++;
02:0177     this._event('upgrade', { key, level: this.state.upgrades[key], cost }); return { ok: true, key,
02:0177+ level: this.state.upgrades[key], cost };
02:0178 }
02:0179 evolve() {
02:0180     const next = CONFIG.machines[this.state.machine + 1]; if (!next) return fail('max-machine');
02:0181     if (this.state.orderIndex < next.requiredOrders) return fail('orders-required');
02:0182     if (this.state.coins < next.cost) return fail('not-enough-coins');
02:0183     this.state.coins -= next.cost; this.state.machine++;
02:0184     this._event('evolve', { machine: this.state.machine, name: next.name }); return { ok: true,
02:0184+ machine: this.state.machine };

```

src/core.js:185 - src/core.js:226

```

02:0185     }
02:0186     _order() {
02:0187         const s = this.state, isLoop = s.orderIndex >= 20;
02:0188         const order = isLoop ? {
02:0189             name: ['城市返场订单', '全球甜蜜补货', '金色派对专供', '爆米花王国'][s.loopIndex % 4] + ' · ' +
02:0189+ (s.loopIndex + 1),
02:0190             target: finite(CONFIG.orders[19].target * Math.pow(1.35, Math.min(s.loopIndex + 1, 1000))),
02:0191             reward: finite(CONFIG.orders[19].reward * Math.pow(1.3, Math.min(s.loopIndex, 1000)))
02:0192         } : CONFIG.orders[s.orderIndex];
02:0193         const previousTarget = isLoop ? finite(CONFIG.orders[19].target * Math.pow(1.35,
02:0193+ Math.min(s.loopIndex, 1000))) : (s.orderIndex > 0 ? CONFIG.orders[s.orderIndex - 1].target : 0);
02:0194         return {
02:0195             ...order, index: isLoop ? 20 + s.loopIndex : s.orderIndex, number: isLoop ? s.loopIndex + 1 :
02:0195+ s.orderIndex + 1,
02:0196             progress: Math.min(1, s.totalProduced / order.target), stageProgress: Math.max(0, Math.min(1,
02:0196+ (s.totalProduced - previousTarget) / Math.max(1, order.target - previousTarget))),
02:0197             ready: s.totalProduced >= order.target, isLoop
02:0198         };
02:0199     }
02:0200     claimOrder() { return this._claimOrder(1); }
02:0201     _claimOrder(multiplier, frozenBonus = 0) {
02:0202         const order = this._order(); if (!order.ready) return fail('order-not-ready');
02:0203         const coins = finite(order.reward * multiplier + frozenBonus); this._addCoins(coins);
02:0204         multiplier = coins / order.reward;
02:0205         if (order.isLoop) this.state.loopIndex++; else this.state.orderIndex++;
02:0206         for (const [id, q] of Object.entries(this.state.pendingRewards)) if (q.kind === 'order') delete
02:0206+ this.state.pendingRewards[id];
02:0207         this._event('order', { index: order.index, name: order.name, coins, multiplier, isLoop:
02:0207+ order.isLoop });
02:0208         if (this.state.orderIndex === 20 && !order.isLoop) { this.state.completedAt =
02:0208+ this.state.playedSeconds; this._event('complete', { seconds: this.state.playedSeconds }); }
02:0209         return { ok: true, coins, order, multiplier };
02:0210     }
02:0211     _rewardOffer(kind) {
02:0212         const s = this.state; if (!REWARDS.includes(kind)) return null;
02:0213         const p = this._production(), next = CONFIG.machines[s.machine + 1], order = this._order();
02:0214         let amount = 0, title = '', description = '', reason = '';
02:0215         if (kind === 'turbo') {
02:0216             amount = CONFIG.turboDuration; title = '涡轮增压'; description = '自动生产 3 倍 · 90 秒';
02:0217         } else if (kind === 'order') {
02:0218             amount = order.reward * 2; title = '订单加价'; description = '额外 +' + formatNumber(amount) + '
02:0218+ 金币 · 总奖励 3 倍';
02:0219             if (!order.ready) reason = 'order-not-ready';
02:0220         } else if (kind === 'sponsor') {
02:0221             amount = next ? Math.floor(Math.min(p.baseIncome * 120, next.cost * 0.6)) : 0;
02:0222             title = '设备赞助'; description = '获得 ' + formatNumber(amount) + ' 金币';
02:0223             if (!next) reason = 'max-machine';
02:0224             else if (s.coins >= next.cost) reason = 'already-affordable';
02:0225             else if (s.orderIndex < next.requiredOrders) reason = 'orders-required';
02:0226             else if (amount < 1) reason = 'production-required';

```

src/core.js:227 - src/core.js:266

```

02:0227     } else {
02:0228         amount = s.offline ? s.offline.coins : 0; title = '离线翻倍'; description = '额外 +' +
02:0228+ formatNumber(amount) + ' 金币';
02:0229         if (!s.offline) reason = 'no-offline-reward';
02:0230     }
02:0231     if (!reason && s.playedSeconds < CONFIG.rewardUnlockSeconds) reason = 'intro-first';
02:0232     const cooldown = Math.max(0, CONFIG.rewardCooldownSeconds - (s.playedSeconds - s.lastRewardAt));
02:0233     if (!reason && cooldown > 0) reason = 'reward-cooldown';
02:0234     return { kind, title, description, amount, duration: kind === 'turbo' ? 90 : undefined,
02:0234+ available: !reason, reason, cooldown };
02:0235 }
02:0236 quoteReward(kind) {
02:0237     const offer = this._rewardOffer(kind); if (!offer || !offer.available) return null;
02:0238     const s = this.state, existing = Object.values(s.pendingRewards)[0];
02:0239     if (existing) return existing.kind === kind ? clone(existing) : null;
02:0240     s.rewardSerial++;
02:0241     const id = 'reward:' + s.rewardSerial + ':' + Math.floor(s.playedSeconds * 1000);
02:0242     const quote = { id, kind, title: offer.title, description: offer.description, amount:
02:0242+ offer.amount, duration: offer.duration || 0,
02:0243         orderIndex: s.orderIndex, loopIndex: s.loopIndex, offlineId: s.offline ? s.offline.id : null,
02:0243+ machine: s.machine, createdAt: s.playedSeconds };
02:0244     s.pendingRewards[id] = quote; return clone(quote);
02:0245 }
02:0246 cancelReward(id) {
02:0247     if (!has(this.state.pendingRewards, id)) return fail('unknown-reward');
02:0248     delete this.state.pendingRewards[id]; return { ok: true };
02:0249 }
02:0250 applyReward(id) {
02:0251     const s = this.state;
02:0252     if (s.claimedRewards.includes(id)) return fail('already-claimed');
02:0253     if (!has(s.pendingRewards, id)) return fail('unknown-reward');
02:0254     const q = s.pendingRewards[id];
02:0255     if (q.kind === 'order' && (q.orderIndex !== s.orderIndex || q.loopIndex !== s.loopIndex ||
02:0255+ !this._order().ready)) { delete s.pendingRewards[id]; return fail('stale-order'); }
02:0256     if (q.kind === 'offline' && (!s.offline || q.offlineId !== s.offline.id)) { delete
02:0256+ s.pendingRewards[id]; return fail('stale-offline'); }
02:0257     delete s.pendingRewards[id]; s.claimedRewards.push(id); s.claimedRewards =
02:0257+ s.claimedRewards.slice(-128);
02:0258     s.lastRewardAt = s.playedSeconds; s.rewardedCount++;
02:0259     let coins = 0;
02:0260     if (q.kind === 'turbo') s.boostSeconds = Math.min(86400, s.boostSeconds + q.duration);
02:0261     else if (q.kind === 'order') { const result = this._claimOrder(1, q.amount); coins = result.coins;
02:0261+ }
02:0262     else if (q.kind === 'sponsor') { this._addCoins(q.amount); coins = q.amount; }
02:0263     else if (q.kind === 'offline') { const result = this._claimOffline(); this._addCoins(q.amount);
02:0263+ coins = result.coins + q.amount; }
02:0264     this._event('reward', { kind: q.kind, amount: q.amount, coins, id }); return { ok: true, kind:
02:0264+ q.kind, amount: q.amount, coins, duration: q.duration };
02:0265 }
02:0266 claimOffline() { return this._claimOffline(); }

```

src/core.js:267 - src/core.js:300

```

02:0267   _claimOffline() {
02:0268       const offline = this.state.offline; if (!offline) return fail('no-offline-reward');
02:0269       this.state.offline = null; this.state.totalProduced = finite(this.state.totalProduced +
02:0269+ offline.production); this._addCoins(offline.coins);
02:0270       for (const [id, q] of Object.entries(this.state.pendingRewards)) if (q.kind === 'offline') delete
02:0270+ this.state.pendingRewards[id];
02:0271       this._event('offline', { coins: offline.coins, amount: offline.production, seconds:
02:0271+ offline.seconds }); return { ok: true, coins: offline.coins, amount: offline.production, seconds:
02:0271+ offline.seconds };
02:0272   }
02:0273   setSetting(key, value) {
02:0274       if (!['sound', 'haptics'].includes(key) || typeof value !== 'boolean') return
02:0274+ fail('invalid-setting');
02:0275       this.state.settings[key] = value; return { ok: true };
02:0276   }
02:0277   _tutorial() {
02:0278       const s = this.state;
02:0279       if (s.taps < 5) return { step: 0, title: '轻点机器，开始爆香', text:
02:0279+ '点一下就生产并出售一份爆米花。先试试点 5 下!', action: 'tap' };
02:0280       if (s.upgrades.tap < 1) return { step: 1, title: '让每次点击更有分量', text:
02:0280+ '在下方升级「爆裂玉米」，每点一下都能生产更多。', action: 'upgrade-tap' };
02:0281       if (s.upgrades.auto < 1) return { step: 2, title: '请一团火焰帮你工作', text:
02:0281+ '升级「自动火力」，松开手也能持续生产、自动赚金币。', action: 'upgrade-auto' };
02:0282       if (s.bursts < 1) return { step: 3, title: '爆锅倒计时', text:
02:0282+ '继续点击积攒能量，满格会免费触发爆米花大喷发!', action: 'tap' };
02:0283       if (s.orderIndex < 1) return { step: 4, title: '签收你的第一笔订单', text:
02:0283+ '累计产量已经达标。领取订单，拿到额外金币奖励。', action: 'order' };
02:0284       return null;
02:0285   }
02:0286   getView() {
02:0287       const s = this.state, next = CONFIG.machines[s.machine + 1] || null;
02:0288       return {
02:0289           state: s, machine: CONFIG.machines[s.machine], nextMachine: next,
02:0290           upgrades: KEYS.map(key => ({ key, name: CONFIG.upgrades[key].name, description:
02:0290+ CONFIG.upgrades[key].description,
02:0291               level: s.upgrades[key], maxLevel: CONFIG.maxUpgradeLevel, cost: this._upgradeCost(key),
02:0291+ canBuy: s.upgrades[key] < CONFIG.maxUpgradeLevel && s.coins >= this._upgradeCost(key) })),
02:0292           production: this._production(), order: this._order(), energy: s.energy, energyMax: 100,
02:0292+ boostSeconds: s.boostSeconds, tutorial: this._tutorial(),
02:0293           canEvolve: !!next && s.orderIndex >= next.requiredOrders && s.coins >= next.cost,
02:0294           evolveReason: !next ? 'max-machine' : s.orderIndex < next.requiredOrders ? 'orders-required' :
02:0294+ s.coins < next.cost ? 'not-enough-coins' : '',
02:0295           completed: s.orderIndex >= 20, offline: s.offline, loadWarning: this.loadWarning,
02:0296           rewards: Object.fromEntries(REWARDS.map(kind => [kind, this._rewardOffer(kind)])),
02:0297           stats: { totalProduced: s.totalProduced, totalCoins: s.totalCoins, playedSeconds:
02:0297+ s.playedSeconds, taps: s.taps, bursts: s.bursts,
02:0298               orders: s.orderIndex, loopOrders: s.loopIndex, rewardedCount: s.rewardedCount, completedAt:
02:0298+ s.completedAt }
02:0299       };
02:0300   }

```

src/core.js:301 - src/platform.js:37

```
02:0301   drainEvents() { const result = this.events; this.events = []; return result; }
02:0302   exportSave(now = Date.now()) { this.state.savedAt = finite(now, this.now); return clone(this.state);
02:0302+ }
02:0303 }
02:0304 module.exports = { CONFIG, Game, formatNumber };
02:0305
03:0001 'use strict';
03:0002
03:0003 // SDK references checked 2026-09-05. No credentials or live ad IDs are bundled.
03:0004 // https://developer.open-douyin.com/docs/resource/zh-CN/mini-game/develop/api/javascript-api/drawing/
03:0004+ picture/tt-create-canvas
03:0005 // https://developer.open-douyin.com/docs/resource/zh-CN/mini-game/develop/api/javascript-api/foundati
03:0005+ on/system/click-event/tt-on-touch-start
03:0006 // https://developer.open-douyin.com/docs/resource/zh-CN/mini-game/develop/guide/open-ability/ad/incen
03:0006+ tive-ads
03:0007 // https://developer.open-douyin.com/docs/resource/zh-CN/mini-game/develop/api/javascript-api/ads/inte
03:0007+ rstitial-ad/interstitial-ad-notice
03:0008 // https://developer.open-douyin.com/docs/resource/zh-CN/mini-game/develop/guide/open-ability/Introduc
03:0008+ tion-for-tech
03:0009 // https://developer.open-douyin.com/docs/resource/zh-CN/mini-game/develop/api/javascript-api/open-cap
03:0009+ acity/sidebar-capacity/tt-check-scene
03:0010 // https://developer.open-douyin.com/docs/resource/zh-CN/mini-game/develop/api/javascript-api/open-cap
03:0010+ acity/sidebar-capacity/tt-navigate-to-scene
03:0011 const SAVE_KEY = 'little_popcorn_factory_v1';
03:0012
03:0013 function createPlatform() {
03:0014   const root = typeof globalThis !== 'undefined' ? globalThis : GameGlobal;
03:0015   const sdk = typeof tt !== 'undefined' ? tt : root.tt;
03:0016   const isDouyin = !(sdk && typeof sdk.createCanvas === 'function');
03:0017   const config = Object.assign({ appId: '', rewardAdUnitId: '', interstitialAdUnitId: '',
03:0018     allowSimulatedAds: false, analyticsEnabled: false, debug: false }, root.POPCORN_CONFIG || {});
03:0019   const doc = typeof document !== 'undefined' ? document : null;
03:0020   const win = typeof window !== 'undefined' ? window : root;
03:0021   const canvas = isDouyin ? sdk.createCanvas() : createBrowserCanvas(doc);
03:0022   const callbacks = { hide: [], show: [], pointer: [], resize: [] };
03:0023   const analytics = [];
03:0024   const startedAt = Date.now();
03:0025   let lastRewardEnd = -Infinity;
03:0026   let lastInterstitial = -Infinity;
03:0027   let nextInterstitialAttempt = 0;
03:0028   let earlyInterstitialCount = 0;
03:0029   let adBusy = false;
03:0030   let rewardAd = null;
03:0031   let lastStorageError = '';
03:0032   let hidden = false;
03:0033   const sidebar = { supported: false, checking: false, fromSidebar: false };
03:0034   let sidebarShowSubscribed = false;
03:0035   let sidebarShowObserved = false;
03:0036   let sidebarCheck = null;
03:0037   let sidebarNavigation = null;
```

src/platform.js:38 - src/platform.js:86

```
03:0038
03:0039 function subscribe(name, callback) {
03:0040     if (typeof callback !== 'function') return function () {};
03:0041     callbacks[name].push(callback);
03:0042     return function () {
03:0043         const index = callbacks[name].indexOf(callback);
03:0044         if (index >= 0) callbacks[name].splice(index, 1);
03:0045     };
03:0046 }
03:0047
03:0048 function emit(name, value) {
03:0049     callbacks[name].slice().forEach(function (callback) { callback(value); });
03:0050 }
03:0051
03:0052 function getSystemInfo() {
03:0053     let raw = {};
03:0054     if (isDouyin && typeof sdk.getSystemInfoSync === 'function') {
03:0055         try { raw = sdk.getSystemInfoSync() || {}; } catch (_) { /* fallback below */ }
03:0056     }
03:0057     const width = positive(raw.windowWidth || raw.screenWidth || win.innerWidth, 480);
03:0058     const height = positive(raw.windowHeight || raw.screenHeight || win.innerHeight, 840);
03:0059     const pixelRatio = positive(raw.pixelRatio || win.devicePixelRatio, 1);
03:0060     return Object.assign({}, raw, {
03:0061         width: width, height: height, windowWidth: width, windowHeight: height,
03:0062         pixelRatio: pixelRatio,
03:0063         safeArea: raw.safeArea || { left: 0, top: 0, right: width, bottom: height, width: width,
03:0063+ height: height }
03:0064     });
03:0065 }
03:0066
03:0067 function notifyHidden() {
03:0068     if (hidden) return;
03:0069     hidden = true;
03:0070     emit('hide');
03:0071 }
03:0072
03:0073 function notifyShown(value) {
03:0074     if (isDouyin) {
03:0075         // Read every native callback, including the cold-start callback and a
03:0076         // repeated show without hide. Never reuse stale launch options on resume.
03:0077         sidebarShowObserved = true;
03:0078         sidebar.fromSidebar = isSidebarLaunch(value);
03:0079     }
03:0080     // A returning device may have changed size even without a resize API.
03:0081     emit('resize', getSystemInfo());
03:0082     if (!hidden) return;
03:0083     hidden = false;
03:0084     emit('show', value);
03:0085 }
03:0086
```

src/platform.js:87 - src/platform.js:131

```

03:0087   if (isDouyin) {
03:0088       // Register synchronously during game.js initialization, before querying any
03:0089       // optional capability. Missing/broken sidebar support cannot stop gameplay.
03:0090       if (typeof sdk.onShow === 'function') {
03:0091           try { sdk.onShow(notifyShown); sidebarShowSubscribed = true; } catch (_) {}
03:0092       }
03:0093       if (!sidebarShowObserved && typeof sdk.getLaunchOptionsSync === 'function') {
03:0094           try {
03:0095               const launch = sdk.getLaunchOptionsSync();
03:0096               if (!sidebarShowObserved) sidebar.fromSidebar = isSidebarLaunch(launch);
03:0097           } catch (_) { /* onShow remains the authoritative warm-start source */ }
03:0098       }
03:0099       const touchMethods = [['onTouchStart', 'down'], ['onTouchMove', 'move'],
03:0100                           ['onTouchEnd', 'up'], ['onTouchCancel', 'cancel']];
03:0101       touchMethods.forEach(function (pair) {
03:0102           if (typeof sdk[pair[0]] !== 'function') return;
03:0103           sdk[pair[0]](function (event) {
03:0104               const points = event.changedTouches || event.touches || [];
03:0105               points.forEach(function (point) {
03:0106                   emit('pointer', { type: pair[1], x: number(point.screenX, point.clientX),
03:0107                                   y: number(point.screenY, point.clientY), id: point.identifier == null ? 0 :
03:0107+ point.identifier });
03:0108               });
03:0109           });
03:0110       });
03:0111       if (typeof sdk.onHide === 'function') sdk.onHide(notifyHidden);
03:0112       // Feature detection: this event is not present in every mini-game runtime.
03:0113       if (typeof sdk.onWindowResize === 'function') sdk.onWindowResize(function () { emit('resize',
03:0113+ getSystemInfo()); });
03:0114   } else {
03:0115       canvas.style.touchAction = 'none';
03:0116       [['pointerdown', 'down'], ['pointermove', 'move'], ['pointerup', 'up'], ['pointercancel',
03:0116+ 'cancel']].forEach(function (pair) {
03:0117           canvas.addEventListener(pair[0], function (event) {
03:0118               if (event.pointerType === 'mouse' && pair[1] === 'down' && event.button !== 0) return;
03:0119               event.preventDefault();
03:0120               if (pair[1] === 'down' && canvas.setPointerCapture) {
03:0121                   try { canvas.setPointerCapture(event.pointerId); } catch (_) { /* unsupported pointer
03:0121+ capture */ }
03:0122               }
03:0123               const rect = canvas.getBoundingClientRect();
03:0124               emit('pointer', { type: pair[1], x: event.clientX - rect.left,
03:0125                               y: event.clientY - rect.top, id: event.pointerId == null ? 0 : event.pointerId });
03:0126               }, { passive: false });
03:0127           });
03:0128       doc.addEventListener('visibilitychange', function () { if (doc.hidden) notifyHidden(); else
03:0128+ notifyShown(); });
03:0129       win.addEventListener('pagehide', notifyHidden);
03:0130       win.addEventListener('pageshow', notifyShown);
03:0131       win.addEventListener('resize', function () { emit('resize', getSystemInfo()); });

```


src/platform.js:132 - src/platform.js:179

```
03:0132   }
03:0133
03:0134   function getSidebarState() {
03:0135     return Object.assign({}, sidebar);
03:0136   }
03:0137
03:0138   function hasSidebarApi() {
03:0139     return isDouyin && sidebarShowSubscribed && typeof sdk.checkScene === 'function' &&
03:0140       typeof sdk.navigateToScene === 'function';
03:0141   }
03:0142
03:0143   function checkSidebar() {
03:0144     if (sidebarCheck) return sidebarCheck;
03:0145     sidebar.supported = false;
03:0146     if (!hasSidebarApi()) return Promise.resolve(getSidebarState());
03:0147     sidebar.checking = true;
03:0148     let resolveCheck;
03:0149     const pending = new Promise(function (resolve) { resolveCheck = resolve; });
03:0150     sidebarCheck = pending;
03:0151     let settled = false;
03:0152     let timer = null;
03:0153     function finish(supported) {
03:0154       if (settled) return;
03:0155       settled = true;
03:0156       if (timer != null) clearTimeout(timer);
03:0157       sidebar.supported = supported === true;
03:0158       sidebar.checking = false;
03:0159       sidebarCheck = null;
03:0160       resolveCheck(getSidebarState());
03:0161     }
03:0162     try {
03:0163       timer = setTimeout(function () { finish(false); }, 5000);
03:0164       const returned = sdk.checkScene({ scene: 'sidebar',
03:0165         success: function (result) { finish(!(result && result.isExist === true)); },
03:0166         fail: function () { finish(false); } });
03:0167       // The documented API is callback-based; a Promise fulfillment is not
03:0168       // evidence of support. Consume unexpected rejections defensively.
03:0169       if (returned && typeof returned.then === 'function') Promise.resolve(returned).catch(function
03:0169+ () { finish(false); });
03:0170     } catch (_) { finish(false); }
03:0171     return pending;
03:0172   }
03:0173
03:0174   function navigateSidebar() {
03:0175     // Only invoke from an explicit user action. Do not defer navigation behind
03:0176     // an async capability probe, which can lose the runtime's user gesture.
03:0177     if (sidebar.checking) return Promise.resolve({ ok: false, reason: 'checking' });
03:0178     if (!sidebar.supported || !hasSidebarApi()) return Promise.resolve({ ok: false, reason:
03:0178+ 'unavailable' });
03:0179     if (adBusy || sidebarNavigation) return Promise.resolve({ ok: false, reason: 'busy' });
```

src/platform.js:180 - src/platform.js:227

```

03:0180     if (hidden) return Promise.resolve({ ok: false, reason: 'hidden' });
03:0181     let resolveNavigation;
03:0182     const pending = new Promise(function (resolve) { resolveNavigation = resolve; });
03:0183     sidebarNavigation = pending;
03:0184     let settled = false;
03:0185     let timer = null;
03:0186     function finish(ok, reason) {
03:0187         if (settled) return;
03:0188         settled = true;
03:0189         if (timer !== null) clearTimeout(timer);
03:0190         sidebarNavigation = null;
03:0191         // A successful jump only means the sidebar opened, never that the user
03:0192         // returned from it. Only native launch/onShow parameters change that flag.
03:0193         resolveNavigation({ ok: ok, reason: reason });
03:0194     }
03:0195     try {
03:0196         timer = setTimeout(function () { finish(false, 'timeout'); }, 10000);
03:0197         const returned = sdk.navigateToScene({ scene: 'sidebar',
03:0198             success: function () { finish(true, 'navigated'); },
03:0199             fail: function () { finish(false, 'failed'); } });
03:0200         if (returned && typeof returned.then === 'function') Promise.resolve(returned).catch(function
03:0200+ ( ) { finish(false, 'failed'); });
03:0201     } catch ( _ ) { finish(false, 'failed'); }
03:0202     return pending;
03:0203 }
03:0204
03:0205 function track(event, data) {
03:0206     const cleanEvent = String(event || 'unknown').replace(/[^a-zA-Z0-9_]/g, '_').slice(0, 64);
03:0207     const cleanData = {};
03:0208     Object.keys(data || {}).slice(0, 24).forEach(function (key) {
03:0209         const value = data[key];
03:0210         if (typeof value === 'number' && Number.isFinite(value)) cleanData[key] = value;
03:0211         else if (typeof value === 'string') cleanData[key] = value.slice(0, 120);
03:0212         else if (typeof value === 'boolean') cleanData[key] = value ? 1 : 0;
03:0213     });
03:0214     analytics.push({ event: cleanEvent, data: cleanData, time: Date.now() });
03:0215     if (analytics.length > 200) analytics.shift();
03:0216     // Production reporting is an explicit deployment choice; debug events stay in memory by default.
03:0217     if (isDouyin && config.analyticsEnabled === true && typeof sdk.reportAnalytics === 'function') {
03:0218         try { sdk.reportAnalytics(cleanEvent, cleanData); } catch ( _ ) { /* metrics must not interrupt
03:0218+ play */ }
03:0219     }
03:0220 }
03:0221
03:0222 function load() {
03:0223     lastStorageError = '';
03:0224     try {
03:0225         const value = isDouyin ? sdk.getStorageSync(SAVE_KEY) : win.localStorage.getItem(SAVE_KEY);
03:0226         if (value == null || value === '') return null;
03:0227         const parsed = typeof value === 'string' ? JSON.parse(value) : value;

```

src/platform.js:228 - src/platform.js:273

```

03:0228         if (!parsed || typeof parsed !== 'object' || Array.isArray(parsed)) throw new
03:0228+ Error('invalid-save');
03:0229         return parsed;
03:0230     } catch (error) {
03:0231         lastStorageError = String(error && error.message || error);
03:0232         return null;
03:0233     }
03:0234 }
03:0235
03:0236 function save(data) {
03:0237     lastStorageError = '';
03:0238     try {
03:0239         const encoded = JSON.stringify(data);
03:0240         if (typeof encoded !== 'string') throw new Error('invalid-save');
03:0241         if (isDouyin) sdk.setStorageSync(SAVE_KEY, encoded);
03:0242         else win.localStorage.setItem(SAVE_KEY, encoded);
03:0243         return true;
03:0244     } catch (error) {
03:0245         lastStorageError = String(error && error.message || error);
03:0246         return false;
03:0247     }
03:0248 }
03:0249
03:0250 function reward(kind, options) {
03:0251     if (adBusy) return Promise.resolve({ completed: false, reason: 'busy' });
03:0252     if (!isDouyin) {
03:0253         const outcome = options && options.simulate;
03:0254         if (config.allowSimulatedAds !== true) return Promise.resolve({ completed: false, reason:
03:0254+ 'unavailable' });
03:0255         if (['complete', 'cancel', 'fail'].indexOf(outcome) === -1) {
03:0256             return Promise.resolve({ completed: false, reason: 'simulation-choice-required', simulated:
03:0256+ true });
03:0257         }
03:0258         lastRewardEnd = Date.now();
03:0259         const completed = outcome === 'complete';
03:0260         track('ad_simulated', { kind: kind, outcome: outcome });
03:0261         return Promise.resolve({ completed: completed, simulated: true,
03:0262             reason: completed ? 'simulated-complete' : outcome === 'cancel' ? 'cancelled' : 'failed' });
03:0263     }
03:0264     if (!validId(config.appId) || !validId(config.rewardAdUnitId) || typeof sdk.createRewardedVideoAd
03:0264+ !== 'function') {
03:0265         track('ad_unavailable', { kind: kind, placement: 'reward' });
03:0266         return Promise.resolve({ completed: false, reason: 'unavailable' });
03:0267     }
03:0268     adBusy = true;
03:0269     track('ad_request', { kind: kind, placement: 'reward' });
03:0270     return new Promise(function (resolve) {
03:0271         let settled = false;
03:0272         let timer = null;
03:0273         let ad = null;

```

src/platform.js:274 - src/platform.js:319

```

03:0274     function finish(completed, reason, error) {
03:0275         if (settled) return;
03:0276         settled = true;
03:0277         if (timer != null) clearTimeout(timer);
03:0278         if (ad) {
03:0279             if (typeof ad.offClose === 'function') ad.offClose(onClose);
03:0280             if (typeof ad.offError === 'function') ad.offError(onError);
03:0281         }
03:0282         adBusy = false;
03:0283         lastRewardEnd = Date.now();
03:0284         track(completed ? 'ad_complete' : 'ad_incomplete', { kind: kind, reason: reason,
03:0285             code: error && (error.errCode || error.errNo) || 0 });
03:0286         resolve({ completed: completed, reason: reason });
03:0287     }
03:0288     function onClose(result) {
03:0289         // Missing/undefined onClose payload must never grant a reward.
03:0290         finish(!(result && result.isEnded === true), result && result.isEnded === true ? 'completed'
03:0290+ : 'cancelled');
03:0291     }
03:0292     function onError(error) { finish(false, 'failed', error); }
03:0293     try {
03:0294         if (!rewardAd) rewardAd = sdk.createRewardedVideoAd({ adUnitId: config.rewardAdUnitId,
03:0294+ multiton: false });
03:0295         ad = rewardAd;
03:0296         if (!ad || typeof ad.onClose !== 'function' || typeof ad.onError !== 'function' || typeof
03:0296+ ad.show !== 'function') {
03:0297             finish(false, 'unavailable');
03:0298             return;
03:0299         }
03:0300         ad.onClose(onClose);
03:0301         ad.onError(onError);
03:0302         timer = setTimeout(function () {
03:0303             finish(false, 'timeout');
03:0304             try { if (ad && typeof ad.destroy === 'function') ad.destroy(); } catch (_) { /* safe
03:0304+ teardown */ }
03:0305             rewardAd = null;
03:0306         }, 180000);
03:0307         // Douyin recommends show directly, with completion driven only by onClose.
03:0308         Promise.resolve(ad.show()).catch(onError);
03:0309     } catch (error) { finish(false, 'failed', error); }
03:0310 });
03:0311 }
03:0312
03:0313 function interstitial() {
03:0314     const now = Date.now();
03:0315     const elapsed = now - startedAt;
03:0316     // Call only after a settled order group or completed machine transition.
03:0317     // These are product caps, stricter than the SDK's 30s/60s minimums.
03:0318     if (!isDouyin || adBusy || elapsed < 90000 || now - lastInterstitial < 120000 ||
03:0319         now - lastRewardEnd < 60000 || now < nextInterstitialAttempt ||

```

src/platform.js:320 - src/platform.js:367

```
03:0320      (elapsed < 1200000 && earlyInterstitialCount >= 2) ||
03:0321      !validId(config.appId) || !validId(config.interstitialAdUnitId) ||
03:0322      typeof sdk.createInterstitialAd !== 'function') return Promise.resolve(false);
03:0323      adBusy = true;
03:0324      nextInterstitialAttempt = now + 30000;
03:0325      return new Promise(function (resolve) {
03:0326          let settled = false;
03:0327          let shown = false;
03:0328          let showRequested = false;
03:0329          let timer = null;
03:0330          let ad = null;
03:0331          function markShown() {
03:0332              if (shown) return;
03:0333              shown = true;
03:0334              lastInterstitial = Date.now();
03:0335              if (lastInterstitial - startedAt < 1200000) earlyInterstitialCount++;
03:0336              track('ad_impression', { placement: 'interstitial' });
03:0337          }
03:0338          function finish(success) {
03:0339              if (settled) return;
03:0340              settled = true;
03:0341              if (timer !== null) clearTimeout(timer);
03:0342              if (ad) {
03:0343                  if (typeof ad.offClose === 'function') ad.offClose(onClose);
03:0344                  if (typeof ad.offError === 'function') ad.offError(onError);
03:0345                  if (typeof ad.offLoad === 'function') ad.offLoad(onLoad);
03:0346                  // SDK only permits destruction after an interstitial has been shown.
03:0347                  if (shown && typeof ad.destroy === 'function') { try { ad.destroy(); } catch (_) {} }
03:0348              }
03:0349              adBusy = false;
03:0350              resolve(success);
03:0351          }
03:0352          function onClose() { if (settled) return; markShown(); finish(true); }
03:0353          function onError(error) {
03:0354              if (settled) return;
03:0355              track('ad_incomplete', { placement: 'interstitial', code: error && (error.errCode ||
03:0355+ error.errNo) || 0 });
03:0356              finish(false);
03:0357          }
03:0358          function onLoad() {
03:0359              if (settled || showRequested) return;
03:0360              showRequested = true;
03:0361              try { Promise.resolve(ad.show()).then(function () { if (!settled) markShown(); }
03:0361+ ).catch(onError); }
03:0362              catch (error) { onError(error); }
03:0363          }
03:0364          try {
03:0365              ad = sdk.createInterstitialAd({ adUnitId: config.interstitialAdUnitId });
03:0366              if (!ad || typeof ad.show !== 'function' || typeof ad.load !== 'function' ||
03:0367                  typeof ad.onClose !== 'function' || typeof ad.onError !== 'function') { finish(false);
```

src/platform.js:367 - src/platform.js:411

```

03:0367+ return; }
03:0368     ad.onClose(onClose);
03:0369     ad.onError(onError);
03:0370     if (typeof ad.onLoad === 'function') ad.onLoad(onLoad);
03:0371     timer = setTimeout(function () { finish(false); }, 180000);
03:0372     const loading = ad.load();
03:0373     if (loading && typeof loading.then === 'function') loading.then(onLoad).catch(onError);
03:0374     } catch (error) { onError(error); }
03:0375     });
03:0376 }
03:0377
03:0378 function vibrate() {
03:0379     try {
03:0380         if (isDouyin && typeof sdk.vibrateShort === 'function') sdk.vibrateShort({ fail: function () {}
03:0380+ });
03:0381         else if (win.navigator && typeof win.navigator.vibrate === 'function')
03:0381+ win.navigator.vibrate(12);
03:0382     } catch (_) { /* haptics are optional */ }
03:0383 }
03:0384
03:0385 return {
03:0386     canvas: canvas, isDouyin: isDouyin, config: config, load: load, save: save,
03:0387     onHide: function (callback) { return subscribe('hide', callback); },
03:0388     onShow: function (callback) { return subscribe('show', callback); },
03:0389     onPointer: function (callback) { return subscribe('pointer', callback); },
03:0390     onResize: function (callback) { return subscribe('resize', callback); },
03:0391     reward: reward, interstitial: interstitial, track: track, getSystemInfo: getSystemInfo, vibrate:
03:0391+ vibrate,
03:0392     getSidebarState: getSidebarState, checkSidebar: checkSidebar, navigateSidebar: navigateSidebar,
03:0393     getAnalytics: function () { return analytics.map(function (entry) { return Object.assign({},
03:0393+ entry, { data: Object.assign({}, entry.data) }); }); },
03:0394     get lastStorageError() { return lastStorageError; },
03:0395     get adBusy() { return adBusy; }
03:0396 };
03:0397 }
03:0398
03:0399 function isSidebarLaunch(value) {
03:0400     if (!value || typeof value !== 'object' || Array.isArray(value)) return false;
03:0401     // Prefer the official onShow fields when present; do not treat custom query
03:0402     // parameters, scene-shaped strings, or contradictory metadata as a return.
03:0403     if (value.launch_from !== null || value.location !== null) {
03:0404         return value.launch_from === 'homepage' &&
03:0405             (value.location === 'sidebar_card' || value.location === 'homepage_expand');
03:0406     }
03:0407     // Cold-start fallback for hosts whose getLaunchOptionsSync only provides scene.
03:0408     // https://developer.open-douyin.com/docs/resource/zh-CN/mini-game/operation1/user-ops/-retention/si
03:0408+ debar
03:0409     return ['021036', '101036', '181036', '261036'].indexOf(value.scene) !== -1;
03:0410 }
03:0411

```

src/platform.js:412 - src/renderer.js:16

```

03:0412 function createBrowserCanvas(doc) {
03:0413   if (!doc) throw new Error('A browser document or Douyin tt runtime is required.');
```

```

03:0414   let canvas = doc.getElementById('game') || doc.querySelector('canvas');
```

```

03:0415   if (!canvas) {
03:0416     canvas = doc.createElement('canvas');
```

```

03:0417     canvas.id = 'game';
```

```

03:0418     doc.body.appendChild(canvas);
03:0419   }
03:0420   return canvas;
03:0421 }
03:0422
03:0423 function validId(value) {
03:0424   return typeof value === 'string' && value.trim().length > 0 && !/^(YOUR_|REPLACE|TODO|test|placeholder)/i.test(value);
03:0424+ }
03:0425 }
03:0426
03:0427 function positive(value, fallback) { return Number.isFinite(Number(value)) && Number(value) > 0 ?
03:0427+ Number(value) : fallback; }
03:0428 function number(value, fallback) { return Number.isFinite(Number(value)) ? Number(value) :
03:0428+ Number(fallback) || 0; }
03:0429
03:0430 module.exports = { createPlatform: createPlatform, SAVE_KEY: SAVE_KEY };
04:0001 'use strict';
04:0002 const { CONFIG, formatNumber: num } = require('./core');
```

```

04:0003 const C = { ink: '#283e32', muted: '#849084', paper: '#f8f6ed', white: '#fffd7', green: '#366348',
04:0003+ mint: '#deebd5', yellow: '#f4ca58', orange: '#e09a55', line: '#e1e4d7', purple: '#ddd4ef' };
04:0004 const FONT = '"Microsoft YaHei", "PingFang SC", system-ui, sans-serif';
04:0005 const rate = n => n>0&&n<10?Number(n.toFixed(2)).toString():num(n);
04:0006 // Full standard text required before gameplay; not an age-rating or identity system.
04:0007 const HEALTH_ADVISORY = ['抵制不良游戏，拒绝盗版游戏。','注意自我保护，谨防受骗上当。',
04:0007+ '适度游戏益脑，沉迷游戏伤身。','合理安排时间，享受健康生活。'];
04:0008 class Renderer {
04:0009   constructor(ctx) { this.c=ctx; this.zones=[]; this.t=0; this.particles=[]; this.floats=[];
04:0009+ this.flash=0; this.truck=0; }
04:0010   box(x,y,w,h,r=14,fill=C.white,stroke) { const c=this.c; c.beginPath(); c.moveTo(x+r,y); c.arcTo(x+w,
04:0010+ y,x+w,y+h,r);c.arcTo(x+w,y+h,x,y+h,r);c.arcTo(x,y+h,x,y,r);c.arcTo(x,y,x+w,y,r);c.closePath();
04:0010+ if(fill){c.fillStyle=fill;c.fill();}if(stroke){c.strokeStyle=stroke;c.lineWidth=1;c.stroke();} }
04:0011   text(s,x,y,size=14,color=C.ink,weight=400,align='left') { const c=this.c;c.font=`${weight} ${size}
04:0011+ px ${FONT}`;c.fillStyle=color;c.textAlign=align;c.textBaseline='middle';c.fillText(String(s),x,y); }
04:0012   wrap(s,x,y,width,size=14,color=C.muted,line=24) { const c=this.c;c.font=`400 ${size}px ${FONT}`;let
04:0012+ str='',n=0;for(const char of String(s)){if(char==='\\n' || c.measureText(str+char).width>width){
04:0012+ this.text(str,x,y+n*line,size,color);str=char==='\\n'?':char;n++;}else str+=char;}
04:0012+ if(str)this.text(str,x,y+n*line,size,color);return(n+1)*line; }
04:0013   circle(x,y,r,fill,stroke){const c=this.c;c.beginPath();c.arc(x,y,r,0,Math.PI*2);c.fillStyle=fill;
04:0013+ c.fill();if(stroke){c.strokeStyle=stroke;c.stroke();}}
04:0014   line(x,y,xx,yy,color=C.ink,width=2){const c=this.c;c.beginPath();c.moveTo(x,y);c.lineTo(xx,yy);
04:0014+ c.strokeStyle=color;c.lineWidth=width;c.lineCap='round';c.stroke();}
04:0015   hit(x,y,w,h,action) { this.zones.push({x,y,w,h,action}); }
04:0016   button(x,y,w,h,label,action,opts={}) { const fill=opts.disabled?'#e6e8de':opts.fill||C.green;
04:0016+ this.box(x,y,w,h,opts.radius||14,fill,opts.stroke);this.text(label,x+w/2,y+h/2,opts.size||15,
```

src/renderer.js:16 - src/renderer.js:40

```

04:0016+ opts.disabled?'#99a092':opts.color||C.white,opts.weight||650,'center');if(!opts.disabled)this.hit(x,y,
04:0016+ w,h,action); }
04:0017   icon(type,x,y,s=22,color=C.ink) { const c=this.c;c.save();c.translate(x,y);c.scale(s/24,s/24);
04:0017+ c.strokeStyle=color;c.fillStyle=color;c.lineWidth=1.8;c.lineCap='round';c.lineJoin='round';
04:0018   if(type==='coin'){this.circle(12,12,9,'#f8d56e','#d7a642');this.text('•',12,11,20,'#b78723',700,
04:0018+ 'center');}
04:0019   else if(type==='bolt'){c.beginPath();c.moveTo(14,1);c.lineTo(5,14);c.lineTo(11,14);c.lineTo(9,23);
04:0019+ c.lineTo(20,9);c.lineTo(13,9);c.closePath();c.fill();}
04:0020   else if(type==='arrow'){this.line(5,12,19,12,color,2);this.line(14,6,20,12,color,2);this.line(14,
04:0020+ 18,20,12,color,2);}
04:0021   else if(type==='play'){c.beginPath();c.moveTo(8,5);c.lineTo(19,12);c.lineTo(8,19);c.closePath();
04:0021+ c.fill();}
04:0022   else if(type==='gear'){for(let i=0;i<8;i++){c.save();c.translate(12,12);c.rotate(i*Math.PI/4);
04:0022+ c.fillRect(-2,-12,4,5);c.restore();}this.circle(12,12,8,color);this.circle(12,12,3,C.paper);}
04:0023   else if(type==='factory'){this.box(2,11,20,11,2,null,color);this.line(3,11,9,6,color);this.line(9,
04:0023+ 6,9,11,color);this.line(9,11,16,6,color);this.line(16,6,16,11,color);c.fillRect(18,2,3,9);for(let i=0;
04:0023+ i<3;i++)c.fillRect(6+i*5,15,2,3);}
04:0024   else if(type==='chart'){this.line(3,2,3,22,color);this.line(3,22,23,22,color);c.fillRect(7,12,3,
04:0024+ 7);c.fillRect(13,8,3,11);c.fillRect(19,3,3,16);}
04:0025   else if(type==='hand'){this.circle(12,15,7,color);this.box(9,2,5,14,2,color);this.box(5,10,5,9,2,
04:0025+ color);}
04:0026   else if(type==='check'){this.line(4,12,9,17,color,3);this.line(9,17,20,5,color,3);}
04:0027   else if(type==='close'){this.line(6,6,18,18,color);this.line(18,6,6,18,color);}
04:0028   else {this.circle(12,12,8,color);}
04:0029   c.restore();
04:0030 }
04:0031 popcorn(x,y,r=5,angle=0) { const c=this.c;c.save();c.translate(x,y);c.rotate(angle);
04:0031+ this.circle(-r*.5,0,r*.66,'#fff8d5');this.circle(r*.4,-r*.35,r*.7,'#fffce9');this.circle(r*.45,r*.5,
04:0031+ r*.65,'#ffeab0');this.circle(-r*.4,r*.55,r*.58,'#fff5c8');this.circle(-r*.05,r*.1,r*.36,'#f0c867');
04:0031+ c.restore(); }
04:0032 emit(event) {
04:0033   if(event.type==='produce'||event.type==='burst') { const burst=event.type==='burst';const
04:0033+ count=burst?65:event.source==='tap'?5:Math.min(5,Math.ceil(event.amount/10));for(let i=0;
04:0033+ i<count&&this.particles.length<180;i++)this.particles.push({x:240+(Math.random()-0.5)*74,y:390,
04:0033+ vx:(Math.random()-0.5)*(burst?300:90),vy:-40-Math.random()*(burst?300:110),life:1.5+Math.random(),
04:0033+ r:4+Math.random()*5,a:Math.random()*6});if(event.source==='tap')this.floats.push({text:`+${
04:0033+ num(event.amount)}`,x:240+(Math.random()-0.5)*80,y:364,life:.8});if(burst)this.flash=1; }
04:0034   if(event.type==='order'){this.truck=1;this.floats.push({text:'订单发车!',x:240,y:420,life:1.7});}
04:0035   if(event.type==='evolve'){this.flash=1;for(let i=0;i<55&&this.particles.length<180;
04:0035+ i++)this.particles.push({x:50+Math.random()*380,y:260,vx:(Math.random()-0.5)*130,
04:0035+ vy:-80-Math.random()*140,life:2+Math.random(),r:5,a:0});}
04:0036 }
04:0037 update(dt) {this.t+=dt;this.flash=Math.max(0,this.flash-dt*1.5);this.truck=Math.max(0,
04:0037+ this.truck-dt*.45);for(const p of this.particles){p.x+=p.vx*dt;p.y+=p.vy*dt;p.vy+=370*dt;p.life-=dt;
04:0037+ p.a+=dt*2;}this.particles=this.particles.filter(p=>p.life>0&&p.y<545);for(const p of this.floats){
04:0037+ p.y-=35*dt;p.life-=dt;}this.floats=this.floats.filter(p=>p.life>0);}
04:0038 draw(view,ui,dt) {
04:0039   this.update(dt);this.zones=[];const c=this.c;c.clearRect(0,0,480,920);this.box(0,0,480,920,0,
04:0039+ C.paper);
04:0040   if(ui.startup){this.welcome(ui);return;}

```


src/renderer.js:41 - src/renderer.js:74

```

04:0041 // Top bar / account
04:0042 this.text('POPCORN WORKS',24,29,10,C.green,750);this.text('小小爆米花厂',24,60,26,C.ink,800);
04:0043 if(ui.isDouyin&&ui.sidebar&&ui.sidebar.supported){
04:0044     this.box(287,18,120,27,13,C.mint);this.text('侧边栏再来玩',347,31,12,C.green,650,'center');
04:0044+ this.hit(282,12,130,46,'sidebar');
04:0045 }else{this.box(287,18,120,27,13,ui.isDouyin?C.mint:C.purple);this.circle(300,31,3,
04:0045+ ui.isDouyin?C.green:'#9380ae');this.text(ui.isDouyin?'工厂营业中':'试玩 • 激励模拟',311,31,11,C.ink,
04:0045+ 600);}
04:0046 const gearY=ui.isDouyin?41:21; // Keep the native menu capsule clear of the settings button
04:0046+ center.
04:0047 this.box(421,gearY,36,36,12,'#ebece1');this.icon('gear',430,gearY+9,18);this.hit(416,gearY-5,46,
04:0047+ 46,'settings');
04:0048 this.box(24,88,432,69,18,C.white,C.line);this.icon('coin',40,101,28);this.text('可用金币',78,107,
04:0048+ 11,C.muted);this.text(num(view.state.coins),78,132,27,C.ink,800);
04:0049 this.line(240,105,240,141,C.line,1);this.text('自动收益 / 秒',262,107,11,C.muted);this.text(`+${
04:0049+ rate(view.production.auto*view.production.price)}`,262,132,23,C.green,750);
04:0050 this.text('01 / 今日生产',24,181,12,C.ink,700);this.text(`累计 ${num(view.state.totalProduced)}
04:0050+ 份`,456,181,12,C.muted,400,'right');
04:0051 this.order(view);
04:0052 this.factory(view,ui);
04:0053 this.bottom(view,ui);
04:0054 if(view.offline&&view.offline.coins>0){this.text('离线收益待领取 →',26,905,11,C.green,650);
04:0054+ this.hit(24,893,170,26,'offline');}else this.text(ui.saved?'已自动保存':'本地存档',26,905,10,C.muted);
04:0054+ this.text('点击生产 / 放置也有收获',454,905,10,C.muted,400,'right');
04:0055 if(ui.modal)this.modal(view,ui);
04:0056 if(ui.toast){const y=ui.modal?174:550;this.box(36,y,408,44,13,'#2d4535');this.text(ui.toast,240,
04:0056+ y+22,12,C.white,600,'center');}
04:0057 }
04:0058 welcome(ui) {
04:0059     this.box(24,28,432,864,28,C.mint);
04:0060     this.text('POPCORN WORKS',240,83,12,C.green,750,'center');
04:0061     this.text('小小爆米花厂',240,129,34,C.ink,800,'center');
04:0062     this.text('从一口小锅, 开始今天的收获',240,172,15,C.green,500,'center');
04:0063     this.circle(240,263,58,'#edf1dd');
04:0064     this.popcorn(240,253,29,-.1);this.popcorn(181,280,10,.3);this.popcorn(299,229,12,-.4);
04:0065     this.box(44,349,392,302,23,C.white);
04:0066     this.text('健康游戏忠告',240,394,27,C.ink,750,'center');
04:0067     HEALTH_ADVISORY.forEach((line,i)=>this.text(line,240,453+i*42,20,C.ink,500,'center'));
04:0068     let y=677;
04:0069     for(const line of ui.publication||[]){y+=this.wrap(line,54,y,372,12,C.green,17)+3;}
04:0070     this.button(64,786,352,57,'开始经营','start',{size:20});
04:0071     this.text('小小爆米花厂 • 1.0.0',240,863,12,C.green,500,'center');
04:0072 }
04:0073 order(v) {const o=v.order;this.box(24,198,432,82,18,'#e9eddf');this.box(37,213,45,45,12,C.white);
04:0073+ this.icon(o.ready?'check':'factory',48,224,23,C.green);this.text(o.isLoop?'循环订单 ${
04:0073+ v.state.loopIndex+1}`: `订单 ${String(v.state.orderIndex+1).padStart(2,'0')} / 20`,94,217,10,C.muted,
04:0073+ 650);this.text(o.name,94,237,15,C.ink,750);
04:0074     this.box(94,254,228,5,2,'#d5ddca');this.box(94,254,Math.max(5,228*Math.min(1,o.progress)),5,2,
04:0074+ C.green);this.text(o.ready?'可以发车':`${num(v.state.totalProduced)} / ${num(o.target)}`,439,265,10,
04:0074+ C.muted,500,'right');this.button(340,211,102,34,o.ready?'装车结算':'查看订单','order',{

```

src/renderer.js:74 - src/renderer.js:97

```

04:0074+ fill:o.ready?C.yellow:C.white,color:C.ink,size:12)); }
04:0075     factory(v,ui) {
04:0076         const c=this.c;this.box(24,293,432,285,22,'#e1ebd8');c.save();this.box(24,293,432,285,22,null);
04:0076+ c.clip();
04:0077         for(let x=40;x<470;x+=36){this.line(x,305,x,568,'#d4e0cb',.6);}for(let y=308;y<580;
04:0077+ y+=36)this.line(28,y,452,y,'#d4e0cb',.6);
04:0078         this.circle(374,334,70,'#e9f0e3');this.circle(386,326,48,'#f1f4e9');this.line(31,520,450,520,
04:0078+ '#bacdb1',2);
04:0079         this.box(42,309,127,26,13,'#f5f6e8');this.circle(55,322,3,C.green);this.text(v.machine.name,66,
04:0079+ 322,11,C.ink,650);
04:0080         if(v.boostSeconds>0){this.box(312,309,128,26,13,C.yellow);this.icon('bolt',321,314,15,C.ink);
04:0080+ this.text(`3 倍增压 ${Math.ceil(v.boostSeconds)}s`,342,322,11,C.ink,600);}else this.text(`STAGE 0${
04:0080+ v.state.machine+1}`,434,322,10,'#82917b',650,'right');
04:0081         // wall shelves and corn sacks
04:0082         this.line(44,416,102,416,'#b6c4a9',5);for(let i=0;i<3;i++){this.box(49+i*17,383,12,31,3,
04:0082+ i%2?'#e8b74e':'#b9cb9e');this.line(52+i*17,390,58+i*17,390,'#f9efc8',2);}
04:0083         this.box(364,470,45,48,7,'#d9c793');this.line(370,479,403,479,'#b09c64',2);this.text('CORN',386,
04:0083+ 497,9,'#8e7c4d',700,'center');this.box(408,484,33,34,6,'#e6d7ac');
04:0084         c.save();c.translate(240,421);this.drawMachine(v.state.machine,
04:0084+ v.production.auto>0||this.particles.length>0);c.restore();
04:0085         // striped popcorn bucket
04:0086         c.save();c.translate(242,504);c.beginPath();c.moveTo(-49,-29);c.lineTo(49,-29);c.lineTo(37,31);
04:0086+ c.lineTo(-37,31);c.closePath();c.fillStyle='#fff9e8';c.fill();c.save();c.clip();for(let i=-40;i<50;
04:0086+ i+=23){c.save();c.rotate(.06);c.fillStyle='#d89465';c.fillRect(i,-30,12,64);c.restore();}c.restore();
04:0086+ this.box(-51,-32,102,8,4,'#fffbee');
04:0087         const pile=13+Math.min(40,Math.floor(v.state.machine*6+v.energy/7));for(let i=0;i<pile;i++){const
04:0087+ x=Math.sin(i*31.2)*39,y=-29-Math.abs(Math.cos(i*11.8))*16-(i%5);this.popcorn(x,y,5.5,i);}this.box(-20,
04:0087+ -6,40,22,9,'#fff9e7');this.text('POP!',0,5,12,C.ink,900,'center');c.restore();
04:0088         for(const p of this.particles){c.globalAlpha=Math.min(1,p.life*2);this.popcorn(p.x,p.y,p.r,p.a);}
04:0088+ c.globalAlpha=1;
04:0089         if(this.truck>0){const x=480-(1-this.truck)*750;this.box(x,474,110,54,8,'#f7c85d');this.box(x+103,
04:0089+ 489,45,39,6,'#658769');this.box(x+111,494,24,15,3,'#dfeadb');this.circle(x+26,532,11,C.ink);
04:0089+ this.circle(x+123,532,11,C.ink);this.circle(x+26,532,5,C.white);this.circle(x+123,532,5,C.white);
04:0089+ this.text('POPCORN',x+55,502,11,C.ink,750,'center');}
04:0090         for(const f of this.floats){c.globalAlpha=Math.min(1,f.life*2);this.text(f.text,f.x,f.y,21,
04:0090+ C.green,800,'center');}c.globalAlpha=1;
04:0091         if(this.flash>0){c.globalAlpha=this.flash*.4;c.fillStyle='#fff7cb';c.fillRect(24,293,432,285);
04:0091+ c.globalAlpha=1;c.restore();
04:0092         this.hit(24,340,432,203,'tap');
04:0093         this.box(44,548,392,19,9,'#cbd9c2');this.box(44,548,Math.max(18,392*v.energy/100),19,9,C.yellow);
04:0093+ this.text(v.energy>80?'快要爆锅了! 继续点击':'免费爆锅能量',57,558,10,C.ink,650);this.text(`${
04:0093+ Math.floor(v.energy)} / 100`,422,558,10,C.ink,650,'right');
04:0094         this.button(24,590,283,55,'点一下, 爆米花!', 'tap',{fill:C.green,size:18});this.icon('hand',45,605,
04:0094+ 24,C.white);this.text(`${num(v.production.tap)} 份`,280,631,10,'#bed5bd',500,'right');
04:0095         this.button(317,590,139,55,v.boostSeconds>0?'续时增压':'涡轮增压','ad:turbo',{fill:C.yellow,
04:0095+ color:C.ink,size:15});this.text('广告 • 3 倍 / 90 秒',387,633,10,'#755b23',500,'center');
04:0096         if(v.tutorial){this.text(`小厂长提示 • ${v.tutorial.title}`,240,663,12,C.green,600,'center');}
04:0096+ else this.text(`每次点击 +${num(v.production.tap*v.production.price)} 金币 • 每份售价 ${
04:0096+ rate(v.production.price)}`,240,663,11,C.muted,400,'center');
04:0097     }

```

src/renderer.js:98 - src/renderer.js:114

```

04:0098   drawMachine(stage,active) {
04:0099       const c=this.c,bob=active?Math.sin(this.t*12)*1.1:0;c.translate(0,bob);if(stage===5)c.scale(.73,
04:0099+   .73);
04:0100       const colors=['#d2a567','#85a898','#b398bf','#7e9bad','#d1956d','#cead4b'];const
04:0100+   color=colors[stage];
04:0101       this.box(-98,31,196,17,6,'#a4b69a');this.box(-83,48,13,37,3,'#8c9f85');this.box(70,48,13,37,3,
04:0101+   '#8c9f85');
04:0102       if(stage===0){this.box(-74,-25,148,58,15,color);this.box(-61,-16,122,28,8,'#f5dfb1');this.box(-80,
04:0102+   -34,160,13,6,'#8c7753');this.box(-33,-43,66,9,4,'#bd9b5b');this.line(77,-9,106,-9,'#78674c',5);
04:0102+   this.line(106,-9,106,-29,'#78674c',5);this.box(98,-37,18,14,6,'#4c6b50');this.box(-19,17,38,15,5,
04:0102+   '#6e7e61');}
04:0103       else if(stage===1){this.box(-73,-61,146,93,18,color);this.box(-59,-47,118,51,10,'#d8e8ce');
04:0103+   this.box(-49,-37,98,33,6,'#f5e5ab');this.box(-82,-69,164,14,6,'#567465');this.circle(43,18,7,
04:0103+   '#f9d36a');this.box(-54,13,61,10,4,'#527161');this.box(-22,30,44,13,4,'#567465');}
04:0104       else if(stage===2){this.box(-98,-28,196,61,12,color);for(const x of [-49,49]){this.box(x-35,-74,
04:0104+   70,68,12,'#ede5ee');this.box(x-41,-83,82,14,5,'#877192');this.box(x-27,-61,54,38,8,'#f4dfa5');
04:0104+   this.circle(x,15,9,'#6f647e');this.circle(x,15,4,C.yellow);}this.box(-18,30,36,17,4,'#877192');}
04:0105       else if(stage===3){this.box(-103,-66,206,99,13,color);this.box(-112,-76,224,14,5,'#547183');
04:0105+   for(const x of [-83,-50,-17,16,49,82]){this.box(x-13,-53,26,54,6,'#dfe9e8');this.box(x-9,-47,18,36,4,
04:0105+   '#f4dfad');this.box(x-5,3,10,19,3,'#4f6c7d');}this.box(-86,26,172,12,5,'#536e7a');}
04:0106       else if(stage===4){this.box(-124,-55,95,86,10,color);this.box(-113,-43,73,40,6,'#f1ddb3');
04:0106+   this.box(-132,-66,111,15,6,'#997057');this.box(-19,-84,111,80,12,'#e7c9a0');this.box(-29,-92,131,14,5,
04:0106+   '#a47859');this.box(-3,-70,80,40,6,'#fff0c8');this.box(-125,17,257,22,9,'#61785e');for(let i=0;i<12;
04:0106+   i++){const x=-115+i*21+(this.t*28%21);this.line(x,22,x-8,34,'#a8bba0',2);}this.box(30,-4,24,32,4,
04:0106+   '#b38c68');}
04:0107       else {this.box(-91,-36,182,67,15,'#c7ad58');this.box(-62,-85,124,53,13,'#e8cd71');this.box(-40,
04:0107+   -133,80,52,12,'#f3d889');this.box(-99,-45,198,13,5,'#9b8849');this.box(-70,-94,140,13,5,'#a78c47');
04:0107+   this.box(-46,-141,92,12,5,'#b69a4f');for(let i=0;i<3;i++){this.box(-68+i*53,-23,30,37,7,'#f7ecbd');
04:0107+   this.circle(-53+i*53,-4,6,'#b39643');}this.box(-27,-70,54,25,7,'#fff0c6');this.circle(0,-111,12,
04:0107+   '#fff7d7');this.text('★',0,-111,16,'#b68c2d',700,'center');this.line(0,-144,0,-168,'#9d8649',3);
04:0107+   this.box(1,-166,31,16,2,'#d88964');}
04:0108       const kernels=stage===0?9:12+stage*3;for(let i=0;i<kernels;i++){const
04:0108+   x=Math.sin(i*3.4+this.t*(active?2:0))*48;this.popporn(x,-29-Math.abs(Math.cos(i*5.8+this.t))*9,4,i);}
04:0109       if(active){for(let i=0;i<3;i++){c.globalAlpha=.22;this.circle(-25+i*24,-103-(this.t*16+i*18)%45,
04:0109+   7+(this.t*3+i)%8,'#ffffff');}c.globalAlpha=1;}
04:0110   }
04:0111   bottom(v,ui) {
04:0112       const tabs=[['upgrades','bolt','设备升级'],['machines','factory','工厂蓝图'],['stats','chart',
04:0112+   '经营记录']];tabs.forEach(([id,icon,label],i)=>{const x=24+i*147;this.box(x,684,138,38,12,
04:0112+   ui.tab===id?C.mint:'#eeeeee5');this.icon(icon,x+15,694,17,ui.tab===id?C.green:C.muted);this.text(label,
04:0112+   x+43,703,12,ui.tab===id?C.green:C.muted,650);this.hit(x,684,138,38,'tab:'+id)});
04:0113       if(ui.tab==='upgrades'){v.upgrades.forEach((u,i)=>{const y=733+i*52;this.box(24,y,432,46,12,
04:0113+   C.white,C.line);this.box(34,y+7,32,32,9,[C.mint,C.purple,'#f7e7b1'][i]);this.icon(['hand','bolt',
04:0113+   'coin'][i],41,y+14,18);this.text(u.name,77,y+14,13,C.ink,700);this.text(`Lv.${u.level}    ${
04:0113+   u.description}`,77,y+33,10,C.muted);this.button(329,y+7,116,32,u.level>=u.maxLevel?'已满级':num(u.cost
04:0113+   )+'  升级','upgrade:'+u.key,{disabled:!u.canBuy,fill:C.green,size:12});});}
04:0114       else if(ui.tab==='machines'){this.box(24,733,432,149,16,C.white,C.line);const next=v.nextMachine;
04:0114+   this.text(next?'下一站, 工厂长大了': '六阶工厂  ·  已全部落成',42,754,12,C.muted);
04:0114+   this.text(next?next.name: '巨型爆米花塔',42,782,23,C.ink,800);this.wrap(next?next.description: '每一粒小小
04:0114+   玉米, 都成了今天的盛大收获。继续完成循环订单, 刷新生产纪录。',42,807,252,11,C.muted,18);if(next){

```

src/renderer.js:114 - src/renderer.js:123

```
04:0114+ this.button(324,751,115,42,'查看换代','machine',{fill:C.yellow,color:C.ink,size:13});this.text(`${
04:0114+ num(next.cost)} 金币`,381,813,11,C.muted,500,'center');this.text(`订单 ${v.state.orderIndex}/${
04:0114+ next.requiredOrders}`,381,834,11,C.muted,500,'center');}else{this.button(324,759,115,42,'竣工纪念',
04:0114+ 'completion',{fill:C.yellow,color:C.ink,size:13});}this.button(42,854,394,20,'查看全部 6 个阶段 →',
04:0114+ 'blueprint',{fill:C.white,color:C.green,size:11});}
04:0115     else {this.box(24,733,432,149,16,C.white,C.line);const stats=[['总生产',
04:0115+ num(v.state.totalProduced)+' 份'],['完成订单',String(v.state.orderIndex+v.state.loopIndex)+' 单'],
04:0115+ ['免费爆锅',String(v.state.bursts)+' 次'],['累计营业',num(v.state.totalCoins)+' 金币']];
04:0115+ stats.forEach(([label,value],i)=>{const x=44+(i%2)*217,y=756+Math.floor(i/2)*67;this.text(label,x,y,
04:0115+ 11,C.muted);this.text(value,x,y+26,19,C.ink,750)});}
04:0116   }
04:0117   modal(v,ui) {
04:0118     const m=ui.modal,c=this.c;this.zones=[];c.fillStyle='rgba(31,48,36,.45)';c.fillRect(0,0,480,920);
04:0118+ const tall=m.type==='blueprint',settings=m.type==='settings',y=tall?133:settings?176:239,
04:0118+ h=tall?666:settings?568:446;this.box(24,y,432,h,24,C.white);this.box(423,y+14,22,22,8,'#efeee4');
04:0118+ this.icon('close',423,y+14,22,C.muted);if(!ui.adBusy)this.hit(410,y+5,42,42,'close');
04:0119     const title=(s,sub)=>{this.text(s,48,y+44,24,C.ink,800);if(sub)this.wrap(sub,48,y+82,378,13,
04:0119+ C.muted,22)};
04:0120     if(m.type==='order'){const o=v.order;title(o.ready?'这一车,满载而归':o.name,
04:0120+ o.ready?'爆米花已装好,选择本次订单的结算方式。': '持续生产即可完成订单,生产出的爆米花会自动出售。');
04:0120+ this.box(48,y+130,384,103,16,'#f2eedb');this.text(o.name,68,y+155,14,C.ink,650);this.text(`基础奖励
04:0120+ ${num(o.reward)} 金币`,68,y+190,23,C.ink,750);this.text(`${num(v.state.totalProduced)} / ${
04:0120+ num(o.target)} 份`,68,y+217,11,C.muted);if(o.ready){this.button(48,y+255,384,53,`广告加价 • 共 ${
04:0120+ num(o.reward*3)} 金币`,`ad:order',{fill:C.yellow,color:C.ink});this.button(48,y+321,384,49,`直接装车
04:0120+ • ${num(o.reward)} 金币`,`claimOrder',{fill:C.green});this.text('广告未完成时,仍可直接领取基础奖励',
04:0120+ 240,y+397,11,C.muted,400,'center');}else{this.button(48,y+310,384,50,'回到工厂,继续生产','close');
04:0120+ this.text(`进度 ${Math.min(100,o.progress*100).toFixed(0)}%`,240,y+270,20,C.green,700,'center');}}
04:0121     else if(m.type==='machine'){const n=v.nextMachine;if(!n){title('全阶段设备已落成',
04:0121+ '继续生产,刷新你的工厂纪录。');this.button(48,y+315,384,50,'回到工厂','close');return;}title(n.name,
04:0121+ n.description);this.box(48,y+130,384,103,16,C.mint);this.text('设备换代条件',68,y+153,12,C.muted);
04:0121+ this.text(`${num(n.cost)} 金币`,68,y+184,25,C.ink,750);this.text(`已完成 ${v.state.orderIndex} / ${
04:0121+ n.requiredOrders} 个主线订单`,68,y+215,12,C.green);this.button(48,y+255,384,53,
04:0121+ v.canEvolve?'换代,开动新机器!':(v.evolveReason==='orders-required'? '先完成所需订单': '金币不足,继续生产'
04:0121+ ),'evolve',{disabled:!v.canEvolve});this.button(48,y+322,384,49,'广告 • 获取设备赞助','ad:sponsor',{
04:0121+ fill:C.yellow,color:C.ink,disabled:v.state.coins>=n.cost});
04:0121+ this.text('永久升级保留,产量与售价随设备一起成长',240,y+400,11,C.muted,400,'center');}
04:0122     else if(m.type==='offline'){const o=v.offline;title('欢迎回来,小厂长',
04:0122+ o?'你离开的时间里,工厂也在认真营业。': '离线收益已经领取。');if(o){this.box(48,y+131,384,98,16,C.mint);
04:0122+ this.text(`离线营业 ${this.duration(o.seconds)}`,68,y+157,13,C.green);this.text(`+${num(o.coins)}
04:0122+ 金币`,68,y+195,29,C.ink,800);this.button(48,y+253,384,53,`广告翻倍 • 共 ${num(o.coins*2)} 金币`,
04:0122+ 'ad:offline',{fill:C.yellow,color:C.ink});this.button(48,y+319,384,50,'领取收益,开工!',
04:0122+ 'claimOffline');this.text('最多结算 8 小时,按永久自动产能的 50% 计算',240,y+400,11,C.muted,400,
04:0122+ 'center');}}
04:0123     else if(m.type==='reward'){const q=m.quote;title('本次激励奖励',q.title);this.box(48,y+126,384,99,
04:0123+ 16,'#f4e9c5');const s=q.kind==='turbo'? '自动产速 ×3,持续 90 秒':(q.kind==='order' || q.kind==='offline'
04:0123+ '? 额外 +${num(q.amount)} 金币': '奖励 +${num(q.amount)} 金币');this.text(s,240,y+162,22,C.ink,750,
04:0123+ 'center');this.text(q.kind==='order'? '本单共领取 ${num(q.amount*1.5)}
04:0123+ 金币': q.kind==='offline'? '本次共领取 ${num(q.amount*2)} 金币': '完成后按当前显示的奖励结算',240,y+199,12,
04:0123+ C.muted,400,'center');if(ui.isDouyin){this.button(48,y+255,384,53,ui.adBusy?'广告加载 /
04:0123+ 播放中...': '观看广告,领取奖励','watch',{disabled:ui.adBusy,fill:C.yellow,color:C.ink});}else{
```

src/renderer.js:123 - src/renderer.js:137

```
04:0123+ this.text('浏览器模拟体验 · 不会播放真实广告',240,y+247,12,C.green,650,'center');this.button(48,y+274,
04:0123+ 384,50,'模拟完整观看 → 领取奖励',{fill:C.yellow,color:C.ink,disabled:ui.adBusy});
04:0123+ this.button(48,y+336,185,42,'模拟中途关闭',{fill:'#ecee3',color:C.ink,size:12,
04:0123+ disabled:ui.adBusy});this.button(247,y+336,185,42,'模拟加载失败',{fill:'#ecee3',
04:0123+ color:C.ink,size:12,disabled:ui.adBusy});}this.text('可随时关闭; 未完成观看不会发放奖励',240,y+405,11,
04:0123+ C.muted,400,'center');}
04:0124     else if(m.type==='settings'){title('工厂设置','小小爆米花厂 · 1.0.0');this.button(48,y+132,384,47,
04:0124+ `音效                                ${v.state.settings.sound?'已开启':'已关闭'}`,`setting:sound',{fill:C.mint,
04:0124+ color:C.ink});this.button(48,y+191,384,47,`震动反馈                                ${
04:0124+ v.state.settings.haptics?'已开启':'已关闭'}`,`setting:haptics',{fill:C.mint,color:C.ink});
04:0124+ this.button(48,y+250,384,47,'玩法说明','help',{fill:'#eeee5',color:C.ink});this.button(48,y+309,384,
04:0124+ 47,'存档与隐私','privacy',{fill:'#eeee5',color:C.ink});this.button(48,y+368,384,47,'健康游戏忠告',
04:0124+ 'health',{fill:'#eeee5',color:C.ink});if(ui.isDouyin&&ui.sidebar&&ui.sidebar.supported)this.button(48
04:0124+ ,y+427,384,47,'从侧边栏回到工厂','sidebar',{fill:C.mint,color:C.ink});
04:0124+ this.text('进度仅保存于当前设备, 请勿清除应用数据',240,y+526,11,C.muted,400,'center');}
04:0125     else if(m.type==='health'){title('健康游戏忠告','合理安排时间, 休息后再来收获。');
04:0125+ HEALTH_ADVISORY.forEach((line,i)=>this.text(line,240,y+143+i*43,20,C.ink,500,'center'));
04:0125+ this.button(48,y+362,384,49,'回到工厂','close');}
04:0126     else if(m.type==='sidebar'){
04:0127         title('从侧边栏, 再回到工厂',ui.sidebar&&ui.sidebar.fromSidebar?'欢迎回来, 你已从侧边栏进入工厂。':'
04:0127+ 下次来玩, 可以从抖音侧边栏找到工厂。');
04:0128         this.box(48,y+136,384,118,16,C.mint);
04:0129         this.text('抖音首页侧边栏',240,y+174,21,C.green,750,'center');
04:0130         this.text('找到「小小爆米花厂」, 继续今天的经营',240,y+219,14,C.ink,500,'center');
04:0131         this.button(48,y+283,384,49,ui.sidebarBusy?'正在打开…':'去侧边栏看看','visitSidebar',{
04:0131+ disabled:ui.sidebarBusy||!ui.sidebar||!ui.sidebar.supported});
04:0132         this.button(48,y+346,384,43,'继续经营','close',{fill:'#eeee5',color:C.ink});
04:0133     }
04:0134     else if(m.type==='help'){title('一口锅, 也能有大梦想','点击生产 → 免费爆锅 → 完成订单 → 升级换代');
04:0134+ this.wrap('① 点击锅或绿色按钮, 爆米花自动售卖。\\n② 升级自动生产, 松开手也能持续赚钱。\\n③
04:0134+ 能量积满自动爆锅; 升级三条永久产线。\\n④ 完成订单并攒够金币, 解锁下一台机器。\\n⑤ 完成 20
04:0134+ 单后继续循环订单、刷新纪录。',48,y+139,380,14,C.ink,32);this.text('电脑: 空格生产 · 1/2/3 升级 · 0 订单
04:0134+ · M 机器',240,y+329,11,C.muted,400,'center');this.button(48,y+362,384,49,'开工!', 'close');}
04:0135     else if(m.type==='privacy'){title('存档与隐私','进度保存在当前浏览器或小游戏的本地存储中。');
04:0135+ this.wrap('游戏保存金币、设备、订单、音效设置及最后离开时间, 用于恢复进度与计算离线收益。\\n本版无账号、无云存
04:0135+ 档, 不主动上传游戏埋点。抖音广告的数据处理以平台实际说明为准。\\n每 5 秒及离开时自动保存。清理浏览器或应用数据后
04:0135+ , 本地进度可能丢失。',48,y+136,382,14,C.ink,27);this.button(48,y+362,384,49,'知道了','close');}
04:0136     else if(m.type==='blueprint'){title('从小锅, 到大工厂','每一次换代, 产量与售价一起提升。');
04:0136+ CONFIG.machines.forEach((machine,i)=>{const yy=y+128+i*77;this.box(48,yy,384,65,13,
04:0136+ i===v.state.machine?C.mint:'#f2f1e8');this.box(61,yy+14,36,36,11,i<=v.state.machine?machine.color:'#e1
04:0136+ e3d8');this.text(String(i+1).padStart(2,'0'),79,yy+32,14,C.ink,750,'center');this.text(machine.name,
04:0136+ 111,yy+23,15,C.ink,700);this.text(i<=v.state.machine?'已落成':`${num(machine.cost)} 金币 · ${
04:0136+ machine.requiredOrders} 个订单`,111,yy+46,11,C.muted);if(i===v.state.machine)this.text('当前',410,
04:0136+ yy+30,11,C.green,600,'right');});this.button(48,y+604,384,40,'回到工厂','close');}
04:0137     else if(m.type==='completion'){title('小小工厂, 大大梦想','20 个主线订单完成! 每一锅都算数。');
04:0137+ this.box(48,y+126,384,136,18,C.mint);this.text('POPCORN WORKS',240,y+152,12,C.green,750,'center');
04:0137+ this.text('工厂竣工纪念',240,y+193,31,C.ink,800,'center');this.text(`累计生产 ${
04:0137+ num(v.state.totalProduced)} 份 · ${v.state.bursts} 次爆锅`,240,y+238,12,C.green,550,'center');
04:0137+ this.wrap('新的循环订单已经到站。机器还在转, 让下一份纪录更香一点。',48,y+291,380,14,C.muted,25);
04:0137+ this.button(48,y+362,384,49,'继续营业, 刷新纪录','close');}
```

src/renderer.js:138 - src/audio.js:38

```

04:0138     }
04:0139     duration(seconds){const m=Math.floor(seconds/60);return m>=60?`${Math.floor(m/60)} 小时 ${m%60}
04:0139+ 分钟`${m} 分钟`;}
04:0140     actionAt(x,y){for(let i=this.zones.length-1;i>=0;i--){const z=this.zones[i];
04:0140+ if(x>=z.x&&x<=z.x+z.w&&y>=z.y&&y<=z.y+z.h)return z.action;}return null;}
04:0141 }
04:0142 module.exports={Renderer,HEALTH_ADVISORY};
04:0143
04:0144
04:0145
05:0001 'use strict';
05:0002
05:0003 // Original, procedural sound effects. No downloads or third-party audio assets.
05:0004 // Native runtimes without Web Audio safely remain silent; gameplay does not depend on sound.
05:0005 class AudioEngine {
05:0006     constructor() {
05:0007         this.enabled = true;
05:0008         this.context = null;
05:0009         this.lastPop = -Infinity;
05:0010         this.voices = 0;
05:0011         this.unavailable = false;
05:0012         this.native = typeof tt !== 'undefined' && typeof tt.createInnerAudioContext === 'function' ? tt
05:0012+ : null;
05:0013         this.nativeSounds = {};
05:0014         this.lastNativePop = 0;
05:0015     }
05:0016
05:0017     setEnabled(enabled) {
05:0018         this.enabled = !!enabled;
05:0019         if (!this.enabled) for (const audio of Object.values(this.nativeSounds)) { try { audio.stop(); }
05:0019+ catch (_) {} }
05:0020         if (!this.enabled && this.context && typeof this.context.suspend === 'function') {
05:0021             try { Promise.resolve(this.context.suspend()).catch(function () {}); } catch (_) {}
05:0022         }
05:0023     }
05:0024
05:0025     unlock() {
05:0026         if (this.native) return;
05:0027         if (!this.enabled || this.unavailable) return;
05:0028         try {
05:0029             if (!this.context) {
05:0030                 const root = typeof globalThis !== 'undefined' ? globalThis : {};
05:0031                 const Context = root.AudioContext || root.webkitAudioContext;
05:0032                 if (!Context) { this.unavailable = true; return; }
05:0033                 this.context = new Context();
05:0034             }
05:0035             if (this.context.state === 'suspended' && typeof this.context.resume === 'function') {
05:0036                 Promise.resolve(this.context.resume()).catch(function () {});
05:0037             }
05:0038         } catch (_) { this.unavailable = true; }

```

src/audio.js:39 - src/audio.js:80

```

05:0039     }
05:0040
05:0041     play(name) {
05:0042         if (!this.enabled) return;
05:0043         if (this.native) { this.playNative(name); return; }
05:0044         this.unlock();
05:0045         const context = this.context;
05:0046         if (!context || context.state === 'suspended' || context.state === 'closed' || this.voices > 18)
05:0046+ return;
05:0047         const now = context.currentTime;
05:0048         if (name === 'pop' || name === 'tap' || name === 'click') {
05:0049             if (now - this.lastPop < 0.035) return;
05:0050             this.lastPop = now;
05:0051             this.tone(640 + Math.random() * 240, 140, 0.085, 0.07, 'triangle', 0);
05:0052         } else if (name === 'burst' || name === 'boom') {
05:0053             for (let index = 0; index < 7; index++) this.tone(300 + index * 80, 100, 0.12, 0.055,
05:0053+ 'triangle', index * 0.035);
05:0054             this.tone(190, 85, 0.28, 0.08, 'sine', 0);
05:0055         } else if (name === 'upgrade' || name === 'unlock' || name === 'machine') {
05:0056             [440, 554, 659, 880].forEach((frequency, index) => this.tone(frequency, frequency, 0.17, 0.06,
05:0056+ 'sine', index * 0.075));
05:0057         } else if (name === 'order' || name === 'reward' || name === 'coin' || name === 'success') {
05:0058             [659, 880, 1047].forEach((frequency, index) => this.tone(frequency, frequency * 1.04, 0.13,
05:0058+ 0.055, 'sine', index * 0.075));
05:0059         } else if (name === 'complete' || name === 'win') {
05:0060             [523, 659, 784, 1047, 784, 1047].forEach((frequency, index) => this.tone(frequency, frequency,
05:0060+ 0.22, 0.06, 'sine', index * 0.12));
05:0061         } else if (name === 'error' || name === 'deny') {
05:0062             this.tone(160, 120, 0.13, 0.035, 'triangle', 0);
05:0063         } else {
05:0064             this.tone(440, 390, 0.05, 0.025, 'sine', 0);
05:0065         }
05:0066     }
05:0067
05:0068     playNative(name) {
05:0069         // Bundled WAVs, reused contexts; official native API supports package-local paths.
05:0070         // https://developer.open-douyin.com/docs/resource/zh-CN/mini-game/develop/guide/basic-function/au
05:0070+ dio
05:0071         const aliases = { tap:'pop', boom:'burst', unlock:'upgrade', machine:'upgrade', reward:'order',
05:0071+ coin:'order', success:'order', win:'complete', deny:'error', offline:'order' };
05:0072         const key = aliases[name] || (['pop','burst','upgrade','order','complete',
05:0072+ 'error'].includes(name)?name:'click');
05:0073         if (key==='pop' && Date.now()-this.lastNativePop<90) return;
05:0074         if (key==='pop') this.lastNativePop=Date.now();
05:0075         try {
05:0076             if (!this.nativeSounds[key]) {
05:0077                 const audio=this.native.createInnerAudioContext();
05:0078                 audio.src='audio/'+key+'.wav';audio.loop=false;audio.volume=.45;audio.obeyMuteSwitch=true;
05:0079                 if(typeof audio.onError==='function')audio.onError(()=>{});
05:0080                 this.nativeSounds[key]=audio;

```

src/audio.js:81 - tools/build.mjs:17

```

05:0081     }
05:0082     const audio=this.nativeSounds[key];audio.stop();audio.play();
05:0083   } catch (_) { /* Unsupported native audio safely degrades without interrupting play. */ }
05:0084 }
05:0085
05:0086 tone(frequency, endFrequency, duration, volume, type, delay) {
05:0087   const context = this.context;
05:0088   if (!context || this.voices >= 20) return;
05:0089   try {
05:0090     const start = context.currentTime + delay;
05:0091     const oscillator = context.createOscillator();
05:0092     const gain = context.createGain();
05:0093     oscillator.type = type;
05:0094     oscillator.frequency.setValueAtTime(frequency, start);
05:0095     oscillator.frequency.exponentialRampToValueAtTime(Math.max(20, endFrequency), start + duration);
05:0096     gain.gain.setValueAtTime(0.0001, start);
05:0097     gain.gain.exponentialRampToValueAtTime(volume, start + 0.008);
05:0098     gain.gain.exponentialRampToValueAtTime(0.0001, start + duration);
05:0099     oscillator.connect(gain);
05:0100     gain.connect(context.destination);
05:0101     this.voices++;
05:0102     oscillator.onended = () => {
05:0103       this.voices = Math.max(0, this.voices - 1);
05:0104       try { oscillator.disconnect(); gain.disconnect(); } catch (_) {}
05:0105     };
05:0106     oscillator.start(start);
05:0107     oscillator.stop(start + duration + 0.02);
05:0108   } catch (_) { /* unsupported Web Audio nodes degrade without affecting the game */ }
05:0109 }
05:0110 }
05:0111
05:0112 module.exports = { AudioEngine: AudioEngine };
06:0001 import { readFile, writeFile, mkdir } from 'node:fs/promises';
06:0002 import path from 'node:path';
06:0003 import { fileURLToPath } from 'node:url';
06:0004 import { generateAudio } from './audio.mjs';
06:0005 const root = path.resolve(path.dirname(fileURLToPath(import.meta.url)), '..');
06:0006 const modules = new Map();
06:0007 async function collect(file) {
06:0008   const id = path.relative(root, file).replaceAll('\\', '/');
06:0009   if (modules.has(id)) return id;
06:0010   modules.set(id, '');
06:0011   let source = await readFile(file, 'utf8');
06:0012   const requires = [...source.matchAll(/require\(['"](\.\/\.?\/[\^"]+)[']\)/g)];
06:0013   for (const match of requires) {
06:0014     const dependency = path.resolve(path.dirname(file), match[1].endsWith('.js') ? match[1] :
06:0014+ match[1] + '.js');
06:0015     if (!dependency.startsWith(root + path.sep)) throw new Error('Dependency outside project');
06:0016     const dependencyId = await collect(dependency);
06:0017     source = source.replace(match[0], `require(${JSON.stringify(dependencyId)})`);

```



```

06:0018 }
06:0019 modules.set(id, source);
06:0020 return id;
06:0021 }
06:0022 const entry = await collect(path.join(root, 'src/main.js'));
06:0023 const bundle = `/* 小小爆米花厂 v1.0 | generated by npm run build */\n(function(){\n'use strict';
06:0023+ \nconst modules=${JSON.stringify(modules)};\nfunction(module,exports,
06:0023+ require){\n${source}\n}\n}).join(',\n')});\nconst cache={}; function require(id){if(cache[id])return
06:0023+ cache[id].exports;if(!modules[id])throw new Error('Unknown module ' +id);const m=cache[id]={exports:{}};
06:0023+ modules[id](m,m.exports,require);return m.exports;} require(${JSON.stringify(entry)});\n})();\n`;
06:0024 await mkdir(path.join(root, 'build/douyin'), { recursive: true });
06:0025 await mkdir(path.join(root, 'web'), { recursive: true });
06:0026 await writeFile(path.join(root, 'web/game.bundle.js'), bundle);
06:0027 await writeFile(path.join(root, 'build/douyin/game.bundle.js'), bundle);
06:0028 await writeFile(path.join(root, 'build/douyin/game.js'), "require('./config.js');
06:0028+ \nrequire('./game.bundle.js');\n");
06:0029 await writeFile(path.join(root, 'build/douyin/game.json'), JSON.stringify({ deviceOrientation:
06:0029+ 'portrait', showStatusBar: false }, null, 2));
06:0030 let config = {appId:'',rewardAdUnitId:'',interstitialAdUnitId:'',allowSimulatedAds:false};
06:0031 try { config = {...config,...JSON.parse(await readFile(path.join(root, 'config.local.json'), 'utf8'))};
06:0031+ }
06:0032 catch(error) { if(error.code!=='ENOENT')throw new Error('config.local.json 无效: '+error.message); }
06:0033 config.allowSimulatedAds=false;
06:0034 await writeFile(path.join(root, 'build/douyin/config.js'), '(typeof globalThis !== "undefined" ?
06:0034+ globalThis : GameGlobal).POPCORN_CONFIG = '+JSON.stringify(config,null,2)+';\n');
06:0035 await writeFile(path.join(root, 'build/douyin/project.config.json'), JSON.stringify({ description:
06:0035+ '小小爆米花厂', setting: { es6: true, minified: false, urlCheck: true }, appid: config.appId,
06:0035+ projectname: 'tiny-popcorn-factory', compileType: 'game' }, null, 2));
06:0036 await generateAudio(path.join(root, 'build/douyin/audio'));
06:0037 console.log(`Built ${modules.size} modules; ${Buffer.byteLength(bundle)/1024}.toFixed(1)} KiB. Web:
06:0037+ web/ • Douyin: build/douyin/`);
07:0001 // Original PCM sound assets for the native mini-game audio API.
07:0002 import { mkdir, writeFile } from 'node:fs/promises';
07:0003 import path from 'node:path';
07:0004 const sounds={pop:[.10,[760]],burst:[.42,[220,330,440,550,660,770]],upgrade:[.48,[440,554,659,880]],
07:0004+ order:[.36,[659,880,1047]],complete:[.82,[523,659,784,1047,784,1047]],click:[.07,[440]],error:[.14,
07:0004+ [160]]};
07:0005 export async function generateAudio(directory){
07:0006   await mkdir(directory,{recursive:true});
07:0007   for(const [name,[duration,notes]] of Object.entries(sounds)){
07:0008     const rate=22050,count=Math.ceil(rate*duration),buf=Buffer.alloc(44+count*2);
07:0009     buf.write('RIFF',0);buf.writeUInt32LE(36+count*2,4);buf.write('WAVEfmt ',8);buf.writeUInt32LE(16,
07:0009+ 16);buf.writeUInt16LE(1,20);buf.writeUInt16LE(1,22);buf.writeUInt32LE(rate,24);
07:0009+ buf.writeUInt32LE(rate*2,28);buf.writeUInt16LE(2,32);buf.writeUInt16LE(16,34);buf.write('data',36);
07:0009+ buf.writeUInt32LE(count*2,40);
07:0010     for(let i=0;i<count;i++){
07:0011       const t=i/rate,noteLength=duration/notes.length,index=Math.min(notes.length-1,
07:0011+ Math.floor(t/noteLength)),local=t-index*noteLength;
07:0012       const envelope=Math.min(1,local/.004)*Math.max(0,1-local/noteLength)**1.8;
07:0013       const freq=notes[index],phase=2*Math.PI*(freq*local-(name==='pop'?220*local*local:0));

```

tools/audio.mjs:14 - web/index.html:19

```

07:0014     const value=Math.sin(phase)*envelope*.25;
07:0015     buf.writeInt16LE(Math.round(value*32767),44+i*2);
07:0016   }
07:0017   await writeFile(path.join(directory,name+'.wav'),buf);
07:0018 }
07:0019 }
08:0001 import http from 'node:http';
08:0002 import path from 'node:path';
08:0003 import { readFile } from 'node:fs/promises';
08:0004 import { fileURLToPath } from 'node:url';
08:0005 const root = path.resolve(path.dirname(fileURLToPath(import.meta.url)), '../web');
08:0006 const port = Number(process.env.PORT || 4173);
08:0007 const types = { '.html':'text/html; charset=utf-8', '.js':'text/javascript; charset=utf-8',
08:0007+ '.css':'text/css; charset=utf-8', '.svg':'image/svg+xml' };
08:0008 http.createServer(async (req, res) => {
08:0009   try {
08:0010     const url = new URL(req.url, 'http://localhost');
08:0011     const target = path.resolve(root, '.' + decodeURIComponent(url.pathname === '/' ? '/index.html' :
08:0011+ url.pathname));
08:0012     if (!target.startsWith(root + path.sep)) { res.writeHead(403); res.end(); return; }
08:0013     const data = await readFile(target);
08:0014     res.writeHead(200, { 'Content-Type': types[path.extname(target)] || 'application/octet-stream',
08:0014+ 'Cache-Control':'no-store' }); res.end(data);
08:0015   } catch { res.writeHead(404); res.end('Not found'); }
08:0016 }).listen(port, '127.0.0.1', () => console.log(`小小爆米花厂 http://127.0.0.1:${port}`));
09:0001 <!doctype html>
09:0002 <html lang="zh-CN">
09:0003 <head>
09:0004   <meta charset="UTF-8">
09:0005   <meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover,
09:0005+ maximum-scale=1">
09:0006   <meta name="theme-color" content="#edf2e9">
09:0007   <meta name="description" content="从一口手摇锅，到整座爆米花工厂。点击、爆锅、升级，创造属于你的爆米花纪录
09:0007+ 。">
09:0008   <title>小小爆米花厂 • 第一版</title>
09:0009   <style>
09:0010     *{box-sizing:border-box}html,body{margin:0;width:100%;height:100%;overflow:hidden;
09:0010+ background:#e9eee5}
09:0011     body{touch-action:none;overscroll-behavior:none;font-family:system-ui,'Microsoft YaHei',
09:0011+ sans-serif}
09:0012     canvas{display:block;position:fixed;inset:0;width:100%;height:100%;touch-action:none;outline:none}
09:0013     .loading{position:fixed;inset:0;display:grid;place-content:center;text-align:center;color:#264333;
09:0013+ gap:12px;pointer-events:none}
09:0014     .loading strong{font-size:24px}.loading span{font-size:13px;color:#6e7d71}
09:0015     .sr-only{position:absolute;left:-9999px;width:1px;height:1px;overflow:hidden}
09:0016   </style>
09:0017 </head>
09:0018 <body>
09:0019   <div class="loading" id="loading"><strong>小小爆米花厂</strong><span>正在给第一锅玉米加热…</span></div>
09:0019+ >

```

web/index.html:20 - web/index.html:24

```
09:0020    <p class="sr-only">游戏快捷键：空格生产，1、2、3升级，O领取订单，M查看机器，Escape关闭面板。浏览器广告为明
09:0020+ 确标注的模拟体验。</p>
09:0021    <script>globalThis.POPCORN_CONFIG={allowSimulatedAds:true,appId:'',rewardAdUnitId:'',
09:0021+ interstitialAdUnitId:''};</script>
09:0022    <script src="game.bundle.js"></script>
09:0023 </body>
09:0024 </html>
```